

Exercise JA

Q1. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ then $|2\vec{a} + 5\vec{b} + 5\vec{c}|$ is

Ans. 3

Sol. (a) $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$
 $\Rightarrow 6 - 2\Sigma \vec{a} \cdot \vec{b} = 9$
 $\Rightarrow \Sigma \vec{a} \cdot \vec{b} = -\frac{3}{2}$ (1)

$$|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$$

$$\Sigma \vec{a}^2 + 2\Sigma \vec{a} \cdot \vec{b} \geq 0$$

for equality $|\vec{a} + \vec{b} + \vec{c}| = 0$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$$

$$5\vec{b} + 5\vec{c} = -5\vec{a}$$

$$|2\vec{a} + 5\vec{b} + 5\vec{c}| = 3|\vec{a}| = 3$$

Q2. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\vec{i} + 3\vec{j} + 4\vec{k}) = (2\vec{i} + 3\vec{j} + 4\vec{k}) \times \vec{b}$ then a possible value of $(\vec{a} + \vec{b}) \cdot (-7\vec{i} + 2\vec{j} + 3\vec{k})$ is

(A) 0 (B) 3 (C) 4 (D) 8

Ans. C

Sol. $(\vec{a} + \vec{b}) \times (2\vec{i} + 3\vec{j} + 4\vec{k}) = 0$

$$\Rightarrow \vec{a} + \vec{b} = \lambda(2\vec{i} + 3\vec{j} + 4\vec{k})$$

$$|\vec{a} + \vec{b}| = \sqrt{29} \Rightarrow |\lambda| = 1$$

$$\vec{a} + \vec{b} = (2\vec{i} + 3\vec{j} + 4\vec{k})$$

$$(\vec{a} + \vec{b}) \cdot (-7\vec{i} + 2\vec{j} + 3\vec{k})$$

$$= -14 + 6 + 12 = 4$$

Q3. The point P is the intersection of the straight line joining the points Q(2, 3, 5) and R(1, -1, 4) with the plane $5x - 4y - z = 1$. If S is the foot of the perpendicular drawn from the point T(2, 1, 4) to QR, then the length of the line segment PS is

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

Ans. A

Sol. Line QR : $\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-5}{1} = \lambda$

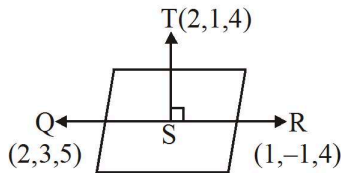
Any point on line QR :

$$(\lambda + 2, 4\lambda + 3, \lambda + 5)$$

$$\therefore \lambda = -\frac{2}{3}$$

$$\therefore \left(\frac{4}{3}, \frac{1}{3}, \frac{13}{3}\right)$$

Also



$$\therefore TQ = TR = \sqrt{5}$$

$\Rightarrow$ S is the mid-point of QR

$$\Rightarrow S\left(\frac{3}{2}, 1, \frac{9}{2}\right) \Rightarrow PS = \frac{1}{\sqrt{2}} \text{ units}$$

Q4. The equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y + z = 3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point $(3, 1, -1)$ is

(A) $5x - 11y + z = 17$ (B) $\sqrt{2}x + y = 3\sqrt{2} - 1$ (C) $x + y + z = \sqrt{3}$

(D) $x - \sqrt{2}y = 1 - \sqrt{2} \therefore$ required plane is $(x + 2y + 3z - 2) + \left(-\frac{7}{2}\right)(x - y + z - 3) = 0 \Rightarrow 5x - 11y + z = 17$

Ans. A

Sol. Let required plane be $(x + 2y + 3z - 2) + \lambda(x - y + z - 3) = 0$

$\therefore$ plane is at a distance $\frac{2}{\sqrt{3}}$ from the point $(3, 1, -1)$

$$\Rightarrow \left| \frac{(3 + 2 - 2) + \lambda(3 - 1 - 1)}{\sqrt{(1 + \lambda)^2 + (2 - \lambda)^2 + (3 + \lambda)^2}} \right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \lambda^2 = \frac{(1 + \lambda)^2 + (2 - \lambda)^2 + (3 + \lambda)^2}{3}$$

$$\Rightarrow 3\lambda^2 = 3\lambda^2 + 2\lambda - 4\lambda^2 + 6\lambda + 14$$

$$\Rightarrow \lambda = -\frac{7}{2}$$

Q5. If the straight line $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$ are coplanar, then the plane(s) containing these two lines is(are)

(A) $y + 2z = -1$ (B) $y + z = -1$ (C) $y - z = -1$ (D) $y - 2z = -1$

Ans. B, C

Sol. $(1, -1, 0); (-1, -1, 0)$

For coplanarity of lines

$$\begin{vmatrix} 2 & 0 & 0 \\ 2 & k & 2 \\ 5 & 2 & k \end{vmatrix} \Rightarrow 2(k^2 - 4) = 0$$

$$\Rightarrow k = 2$$

for $k = 2$

Normal vector $\vec{n} = \hat{j} - \hat{k}$

$$\therefore y - z = -1$$

for $k = -2$

$\therefore$ Required plane : $y + z = \lambda$

$\therefore$ Passes through $(1, -1, 0)$

$$\Rightarrow \lambda = -1$$

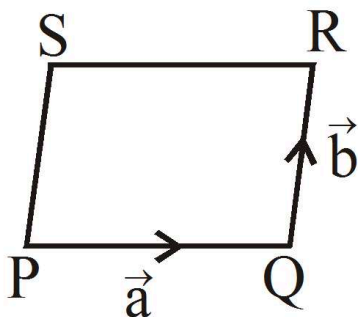
$$\therefore y + z = -1$$

Q6. Let $\vec{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\vec{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\vec{PT}, \vec{PQ}$ and $\vec{PS}$ is

(A) 5 (B) 20 (C) 10 (D) 30

Ans. C

Sol.



$$\vec{a} + \vec{b} = \vec{PR} \quad \& \quad \vec{a} - \vec{b} = \vec{QS}$$

$$\vec{a} = \frac{\vec{PQ} + \vec{QS}}{2} \quad \& \quad \vec{b} = \frac{\vec{PR} - \vec{QS}}{2}$$

$$\vec{a} = 2\hat{i} - \hat{j} - 3\hat{k} \quad \& \quad \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\text{Volume} = \begin{vmatrix} 2 & -1 & -3 \\ 1 & 2 & 1 \\ 0 & 2 & 3 \end{vmatrix}$$

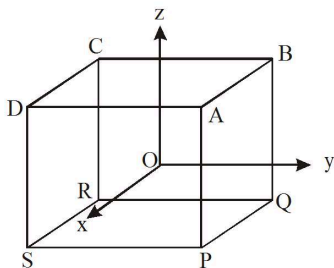
$$2(4) + (3 - 1) - 3(2 - 2)$$

$$8 + 2 = 10$$

Q7. Consider the set of eight vectors $V = \{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is

Ans. 5

Sol.



O is at the centre of cube

ABCDPQRS

The 8 vectors will represent

$$\vec{OA}, \vec{OB}, \dots, \vec{OD}, \vec{OP}, \dots, \vec{OS}$$

any three out of these 8 will be coplanar when two of them are collinear. There are 4 pairs of collinear vectors

$$\vec{OA} \& \vec{OR}, \vec{OB} \& \vec{OS}, \vec{OC} \& \vec{OP}, \vec{OD} \& \vec{OQ}$$

(it will generate $4 \times 6 = 24$ set of coplanar vectors) rest of the combination of 3 vectors will form three edges of a tetrahedron so they will be not coplanar.

So number of non-coplanar vectors

$${}^8C_3 - 4 \cdot 6 = 32$$

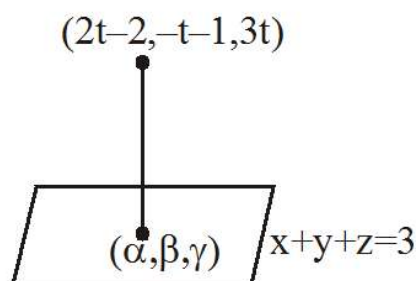
Q8. Perpendiculars are drawn from points on the line $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$ to the plane $x + y + z = 3$. The feet of perpendiculars lie on the line

(A) $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$ (B) $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$ (C) $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$

(D) $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$

Ans. D

Sol. $\frac{\alpha + 2t + 2}{1} = \frac{\beta + t + 1}{1} = \frac{\gamma - 3t}{1} = k$



$$\alpha = k + 2t - 2$$

$$\beta = k - t - 1$$

$$\gamma = k + 3t$$

$$\alpha + \beta + \gamma = 3$$

$$k = \frac{6 - 4t}{3}$$

$$\alpha = \frac{6 - 4t}{3} + 2t - 2 = \frac{2t}{3}$$

$$\beta = \frac{6 - 4t}{3} - t - 1 = \frac{5t + 6}{3}$$

$$\gamma = \frac{6 - 4t}{3} + 3 = \frac{5t + 6}{3}$$

$$\Rightarrow \frac{3\alpha}{2} = \frac{3\beta}{-7} = \frac{3\gamma - 6}{5}$$

$$\Rightarrow \frac{x}{2} = \frac{y-3}{-7} = \frac{z-2}{5}$$

Q9. A line l passing through the origin is perpendicular to the lines

$$l_1 : (3 + t)\hat{i} + (-1 + 2t)\hat{j} + (4 + 2t)\hat{k}, -\infty < t < \infty$$

$$l_2 : (3 + 2s)\hat{i} + (3 + 2s)\hat{j} + (2 + s)\hat{k}, -\infty < s < \infty$$

hen, the coordinate(s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is (are)

$$(A) \left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right) \quad (B) (-1, -1, 0) \quad (C) (1, 1, 1) \quad (D) \left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$$

Ans. B, D

Sol. $l_1 : \vec{r} = (3, -1, 4) + (1, 2, 2)t$

$$l_2 : \vec{r} = (3, 3, 2) + (2, 2, 1)s$$

vector perpendicular to l_1 and l_2 :

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = -2\hat{i} + 3\hat{j} - 2\hat{k}$$

$$\therefore \text{Equation of line } l_1 : \vec{r} = 0 + (-2, 3, -3)\lambda$$

Point of intersection of l_1 and l_2 :

$$3 + t = -2\lambda$$

$$-1 + 2t = 3\lambda$$

$$4 + 2t = -2\lambda$$

On solving we get $\lambda = -1, t = -1$

$\therefore$ Point of intersection of l_1 and l : P(2, -3, 2)

A point on l_2 at distance of $\sqrt{17}$ from P:

$$\Rightarrow (1 + 2s)^2 + (6 + 2s)^2 + s^2 = 17$$

$$\Rightarrow s = -\frac{10}{9}; = -2$$

for above s, point will be (B), (D)

Q10. Two lines $L_1: x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$ and $L_2: x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$ are coplanar. Then a can take value(s)

$$(A) 1 \quad (B) 2 \quad (C) 3 \quad (D) 4$$

Ans. D, A

Sol. $L_1 : \frac{x-5}{0} = \frac{y}{3-\alpha} = \frac{z}{-2}$

$$L_2 : \frac{x-\alpha}{0} = \frac{y}{-1} = \frac{z}{2-\alpha}$$

for lines to be coplanar

$$\begin{vmatrix} 5 - \alpha & 0 & 0 \\ 0 & 3 - \alpha & -2 \\ 0 & -1 & 2 - \alpha \end{vmatrix} = 0$$

$$\Rightarrow (5 - \alpha)((3 - \alpha)(2 - \alpha) - 2) = 0$$

$$\Rightarrow (5 - \alpha)(-\alpha^2 - 5\alpha + 4) = 0$$

$$\Rightarrow \alpha = 1, 4, 5$$

- Q11. Consider the line $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$, $L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$ and the planes $P_1: 7x + y + 2z = 3$, $P_2: 3x + 5y - 6z = 4$. Let $ax + by + cz = d$ be the equation of the plane passing through the point of intersection of lines L_1 and L_2 . and perpendicular to planes P_1 and P_2 .

Match List – I with List – II and select the correct answer using the code given below the lists :

List I	List II
P. $a =$	1. 13
Q. $b =$	2. -3
R. $c =$	3. 1
S. $d =$	4. -2

Codes :

- (A) P Q R S (B) 1 3 4 2 (C) 3 2 1 4 (D) 2 4 1 3
3 2 4 1

Ans. A

Sol. For point of intersection of L_1 and L_2

$$\begin{cases} 2\lambda + 1 = \mu + 4 \\ -\lambda - 3 = \mu + 3 \\ \lambda - 3 = 2 - 3 \end{cases} \Rightarrow \lambda = 1$$

$\Rightarrow$ point of intersection is $(5, -2, -1)$

Now, vector normal to the plane is

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix} = -16(\hat{i} + 3\hat{j} - 2\hat{k})$$

Let equation of required plane be

$$x - 3y - 2z = \alpha$$

$\therefore$ it passes through $(5, -2, -1)$

$$\therefore \alpha = 13$$

$\Rightarrow$ equation of plane is $x - 3y - 2z = 13$

Q12. Match List-I with List-II and select the correct answer using the code given below the lists:

List- I

List-II

P. Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. 1.
100

Then the volume of the parallelepiped determined by vectors $2(\vec{a} \times \vec{b}), 3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$ is

Q. Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. 2. 30

Then the volume of the parallelepiped determined by vectors $3(\vec{a} + \vec{b}), (\vec{b} + \vec{c})$ and $2(\vec{c} + \vec{a})$ is

R. Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is 3. 24

S. Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a} + \vec{b})$ and $\vec{a}$ is 4. 60

Codes:

(A) P Q R S (B) 2 3 1 4 (C) 3 4 1 2 (D) 1 4 3 2
4 2 3 1

Ans. C

Sol. (P) Given $[\vec{a}\vec{b}\vec{c}] = 2$

$$[2(\vec{a} \times \vec{b})3(\vec{b} \times \vec{c})(\vec{c} \times \vec{a})] \\ = 6[\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] = 6[\vec{a}\vec{b}\vec{c}]^2 = 24$$

(Q) Given $[\vec{a}\vec{b}\vec{c}] = 5$

$$\left| \frac{1}{2}(2\vec{a} + 3\vec{b}) \times (\vec{a} - \vec{b}) \right| = \frac{1}{2} |0 + 3\vec{b} \times \vec{a} - 2\vec{a} \times \vec{b}| \\ = \frac{1}{2} |-5\vec{a} \times \vec{b}| = \frac{5}{2} |\vec{a} \times \vec{b}| = \frac{5}{2} \cdot 40 = 100$$

(S) Given $|\vec{a} \times \vec{b}| = 30$

$$|(\vec{a} + \vec{b}) \times \vec{a}| = |0 + \vec{b} \times \vec{a}| = 30$$

Q13. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y} \times \vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z} \times \vec{x}$, then

(A) $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$ (B) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$ (C) $\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$ (D) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{z} - \vec{y})$

Ans. C, B, A

Sol. $\vec{a}$ is in direction of $\vec{x} \times (\vec{y} \times \vec{z})$

i.e. $(\vec{x} \cdot \vec{z})\vec{y} - (\vec{x} \cdot \vec{y})\vec{z}$

$$\Rightarrow \vec{a} = \lambda_1 \left[2 \times \frac{1}{2} (\vec{y} - \vec{z}) \right]$$

$$\vec{a} = \lambda_1 (\vec{y} - \vec{z}) \quad \dots\dots\dots(1)$$

Now $\vec{a} \cdot \vec{y} = \lambda_1 (\vec{y} \cdot \vec{y} - \vec{y} \cdot \vec{z})$

$$= \lambda_1 (2 - 1) \Rightarrow \lambda_1 = \vec{a} \cdot \vec{y} \quad \dots\dots\dots(2)$$

From (1) and (2), $\vec{a} = \vec{a} \cdot \vec{y} (\vec{y} - \vec{z})$

Similarly, $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$

Now, $\vec{a} \cdot \vec{b} = (\vec{a} \cdot \vec{z})(\vec{b} \cdot \vec{z}) \left[(\vec{y} - \vec{z}) \cdot (\vec{y} - \vec{x}) \right]$

$$= (\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z}) [1 - 1 - 2 + 1]$$

$$= -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$$

Q14. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$

If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$, where p, q, and r are scalars, then the value of $\frac{p^2 + 2q^2 + r^2}{q^2}$ is

Ans. 4

Sol. $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = p + q(\vec{a} \cdot \vec{b}) + r(\vec{a} \cdot \vec{c})$$

And $[\vec{a} \vec{b} \vec{c}] = \frac{1}{\sqrt{2}}$

$$p + \frac{q}{2} + \frac{r}{2} [\vec{a} \vec{b} \vec{c}] \quad \dots\dots\dots(1)$$

$$\frac{p}{2} + q + \frac{r}{2} = 0 \quad \dots\dots\dots(2)$$

$$\frac{p}{2} + \frac{q}{2} + r = [\vec{abc}] \dots\dots\dots(3)$$

$$\Rightarrow p = r = -q$$

$$\frac{p^2 + 2q^2 + r^2}{q^2} = 4$$

Q15. List-I

List-II

P. Let $y(x) = \cos(3 \cos^{-1} x)$, $x \in [-1, 1]$, $x \neq \pm \frac{\sqrt{3}}{2}$. Then

1. 1

$$\frac{1}{y(x)} \left\{ (x^2 - 1) \frac{d^2 y(x)}{dx^2} + x \frac{dy(x)}{dx} \right\} \text{ equals}$$

Q. Let $A_1, A_2, \dots, A_n$ ($n > 2$) be the vertices of a regular polygon of

2. 2

n sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point A_k , $k = 1, 2, \dots, n$. If $\left| \sum_{k=1}^{n-1} (\vec{a}_k \times \vec{a}_{k+1}) \right| = \left| \sum_{k=1}^{n-1} (\vec{a}_k \cdot \vec{a}_{k+1}) \right|$, then the minimum value of n is

R. If the normal from the point $P(h, 1)$ on the ellipse $\frac{x^2}{6} + \frac{y^2}{3} = 1$ is

3. 8

perpendicular to the line $x + y = 8$, then the value of h is

S. Number of positive solutions satisfying the equation

4. 9

$$\tan^{-1} \left(\frac{1}{2x+1} \right) + \tan^{-1} \left(\frac{1}{4x+1} \right) = \tan^{-1} \left(\frac{2}{x^2} \right) \text{ is}$$

Codes:

(A) P Q R S (B) 2 4 3 1 (C) 4 3 1 2 (D) 2 4 1 3
4 3 2 1

Ans. A

Sol. (P) $y = \cos(3 \cos^{-1} x)$

$$y' = \frac{3 \sin(3 \cos^{-1} x)}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} y' = 3 \sin(3 \cos^{-1} x)$$

$$\Rightarrow \frac{-x}{\sqrt{1-x^2}} y' + \sqrt{1-x^2} = 3 \cos(3 \cos^{-1} x) \cdot \frac{-3}{\sqrt{1-x^2}}$$

$$\Rightarrow -xy' + (1-x^2) y'' = -9y$$

$$\Rightarrow \frac{1}{y} \left[(x^2 - 1) y'' + xy' \right] = 9$$

$$(Q) (a_k \times a_{k+1}) = r^2 \sin \frac{2\pi}{n}$$

$$a_k \cdot a_{k+1} = r^2 \cos \frac{2\pi}{n}$$

$$\Rightarrow \left| \sum_{k=1}^{n-1} \vec{a}_k \times \vec{a}_{k+1} \right| = \left| \sum_{k=1}^{n-1} a_k \cdot a_{k+1} \right|$$

$$\Rightarrow r^2 (n-1) \sin = \frac{2\pi}{n} r (n-1) \cos \frac{2\pi}{n}$$

$$\Rightarrow \tan \frac{2\pi}{n} = 1 \Rightarrow n = \frac{8}{4k+1}$$

$$(R) \frac{h^2}{6} + \frac{1^3}{3} = 1, h = \pm 2$$

$$\text{Tangent at } (2, 1) \text{ is } \frac{2x}{6} + \frac{y}{3} = 1 \text{ } x + y = 3.$$

$$(S) \tan^{-1} \left(\frac{1}{2x+1} \right) + \tan^{-1} \frac{1}{4x+1} = \tan^{-1} \frac{2}{x^2}$$

$$\tan^{-1} \left(\frac{3x+1}{4x^2+3x} \right) = \tan^{-1} \frac{2}{x^2}$$

$$\Rightarrow 3x^2 - 7x - 6 = 0$$

$$x = -\frac{2}{3}, 3$$

Q16. From a point $p(\lambda, \lambda, \lambda)$, perpendiculars PQ and PR are drawn respectively on the lines $y = x, z = 1$ and $y = -x, z = -1$. If P is such that $\angle QPR$ is a right angle, then the possible value(s) of λ is(are)

$$(A) \sqrt{2} \quad (B) 1 \quad (C) -1 \quad (D) -\sqrt{2}$$

Ans. C

Sol. Line is

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{z-0}{1} = \alpha \quad \dots\dots\dots(1)$$

$$Q(\alpha, \alpha, 1)$$

Direction ratio fo PQ are

$$\lambda - \alpha, \lambda - \alpha - 1$$

Since PQ is perpendicular to (1)

$$\therefore \lambda - \alpha + \lambda - \alpha + 0 = 0$$

$$\lambda = \alpha$$

$\therefore$ Direction ratio of PQ are

$$0, 0, \lambda -$$

Another line is

$$\frac{x-0}{-1} = \frac{y-0}{1} = \frac{z+1}{0} = \beta \quad \dots\dots\dots(2)$$

$$\therefore R(-\beta, \beta, -1)$$

$\therefore$ Direction ratio of PR are

$$\lambda + \beta, -\beta, \gamma + 1$$

Since PQ is perpendicular to (ii)

$$\therefore \beta = 0$$

$$\therefore R(0, 0, -1)$$

and Direction ratio of PQ are $\gamma, \gamma, \gamma, +1$

Since $PQ \perp PR$

$$\therefore 0 + 0 + \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow B, C$$

For $\lambda = 1$ the point is on the line so it will be rejected.

$$\Rightarrow \lambda = -1$$

Q17. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3 . Let the components of a vector $\vec{s}$ along $\vec{p}, \vec{q}$ and $\vec{r}$ be 4, 3 and 5 respectively. If the components of this vector $\vec{s}$ along $(-\vec{p} + \vec{q} + \vec{r})$, $(\vec{p} - \vec{q} + \vec{r})$ and $(-\vec{p} - \vec{q} + \vec{r})$ are x, y and z respectively, then the value of $2x + y + z$ is :

Ans. 9

$$\text{Sol. } \vec{s} = 4\vec{p} + 3\vec{q} + 5\vec{r}$$

$$\vec{s} = x(-\vec{p} + \vec{q} + \vec{r}) + y(\vec{p} - \vec{q} + \vec{r}) + z(-\vec{p} - \vec{q} + \vec{r})$$

$$\vec{s} = (-x + y - z)\vec{p} + (x - y - z)\vec{q} + (x + y + z)\vec{r}$$

$$\Rightarrow -x + y - z = 4$$

$$\Rightarrow x - y - z = 3$$

$$\Rightarrow x + y + z = 5$$

$$\text{On solving we get } x = 4, y = \frac{9}{2}, z = -\frac{7}{2}$$

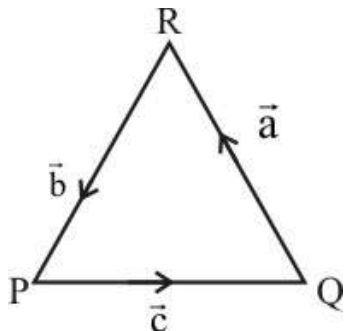
$$\Rightarrow 2x + y + z = 9$$

Q18. Let ΔPQR be a triangle. Let $\vec{a} = \overrightarrow{QR}, \vec{b} = \overrightarrow{PQ}$ and $\vec{c} = \overrightarrow{PR}$. If $|\vec{a}| = 12, |\vec{b}| = 4\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 24$, then which of the following is (are) true ?

(A) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (B) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$ (C) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (D) $\vec{a} \cdot \vec{b} = -72$

Ans. D, C, A

Sol.



$$\vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow \vec{b} + \vec{c} = -\vec{a}$$

$$\Rightarrow 48 + \vec{c}^2 + 48 = 144$$

$$\Rightarrow \vec{c}^2 = 48$$

$$\Rightarrow |\vec{c}|^2 = 4\sqrt{3}$$

$$\therefore \frac{|\vec{c}|^2}{2} - |\vec{a}| = 24 - 12 = 12$$

Further

$$\vec{a} + \vec{b} = -\vec{c}$$

$$\Rightarrow 144 + 48 + 2\vec{a} \cdot \vec{b}$$

$$= 48$$

$$\Rightarrow \vec{a} \cdot \vec{b} = -72$$

$$\therefore \vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 0$$

$$\therefore |\vec{a} \times \vec{b} + \vec{a}| = |\vec{a} \times \vec{b}| = 2 \cdot \sqrt{144 \cdot 48 - (72)^2} = 48\sqrt{3}$$

Q19. In $\mathbb{R}^3$, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2 , which passes through the intersection of P_1 and P_2 . If the distance of the point $(0, 1, 0)$ from P_3 is 1 and the distance of a point (α, β, γ) from P_3 is 2, then which of the following relations is (are) true ?

(A) $2\alpha + \beta + 2\gamma + 2 = 0$ (B) $2\alpha - \beta + 2\gamma + 4 = 0$ (C) $2\alpha + \beta - 2\gamma - 10 = 0$

$$(D) 2\alpha - \beta + 2\gamma - 8 = 0$$

Ans. B, D

Sol. Let P_3 be $P_2 + \lambda P_1 = 0 \Rightarrow x + \lambda y + z - 1 = 0$

Distance from $(0, 1, 0)$ is 1

$$\therefore \frac{0 + \lambda + 0 - 1}{\sqrt{1 + \lambda^2 + 1}} = \pm 1$$

$$\therefore \left| \frac{2\alpha - \beta + 2\gamma - 2}{3} \right| = 2 \quad 2\alpha - \beta + 2\gamma = 2 \pm 6$$

$\therefore$ option (B, D) are correct.

Q20. In R^3 , let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1 : x + 2y - z + 1 = 0$ and $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1 .

Which of the following points lie (s) on M ?

$$(A) \left(0, -\frac{5}{6}, -\frac{2}{3}\right) \quad (B) \left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right) \quad (C) \left(-\frac{5}{6}, 0, \frac{1}{6}\right) \quad (D) \left(-\frac{1}{3}, 0, \frac{2}{3}\right)$$

Ans. A, B

Sol. Let $\vec{v}$ be the vector along L

$$\text{then } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 2 & -1 & 1 \end{vmatrix} = \hat{i} - 3\hat{j} - 5\hat{k}$$

So any point on line L is $A(\lambda - 3\lambda, -5\lambda)$

Foot of perpendicular from A to P_1 is

$$\frac{h - \lambda}{1} = \frac{k - 3\lambda}{-1} = \frac{l + 5\lambda}{-1} = -\frac{(\lambda - 6\lambda + 5\lambda + 1)}{1 + 4 + 1} = -\frac{1}{6}$$

$$h = \lambda - \frac{1}{6}, k = -3\lambda - \frac{1}{3}, l = -5\lambda + \frac{1}{6}$$

$$\text{so foot is } \left(\lambda - \frac{1}{6}, -3\lambda - \frac{1}{3}, 5\lambda + \frac{1}{6}\right)$$

So (A, B)

Q21. Let $\hat{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ be a unit vector in R^3 and $\hat{\omega} = \frac{1}{\sqrt{6}}(\hat{i} + \hat{j} + 2\hat{k})$. Given that there exists a vector $\vec{v}$ in R^3 such that $|\hat{u} \times \vec{v}|$ and $\hat{\omega} \cdot (\hat{u} \times \vec{v}) = 1$. Which of the following statement (s) is (are) correct ?

(A) There is exactly one choice for such $\vec{v}$

(B) There are infinitely many choices for such $\vec{v}$

(C) If $\hat{u}$ lies in the xy-plane than $|u_1| = |u_2|$ (D) If $\hat{u}$ lies in the xz-plane than $2|u_1| = |u_3|$

Ans. B, C

Sol. $\vec{W} \cdot (\hat{u} \times \vec{v}) = 1$

$$\Rightarrow |\vec{W}| |\hat{u} \times \vec{v}| \cos \alpha = 1$$

$$\cos \alpha = 1$$

$$\Rightarrow \hat{w} \perp \hat{u} \text{ and } \hat{w} \perp \vec{v}$$

as it is given there exist a vector $\vec{V}$

$$\Rightarrow \hat{w} \text{ must be } \perp \text{ to } \hat{u}$$

hence infinite many such $\vec{V}$ exists.

$$\text{if } \hat{u} = u_1 \hat{i} + u_2 \hat{j}$$

$$\vec{u} \cdot \vec{w} = 0 \Rightarrow (u_1 + u_2) = 0$$

$$\Rightarrow |u_1| = |u_2|$$

$$\text{if } \mathbf{u} = u_1 \hat{i} + u_3 \hat{k}$$

$$\vec{u} \cdot \vec{w} = 0$$

$$u_1 + 2u_3 = 0$$

$$\Rightarrow |u_1| = 2|u_3|.$$

Q22. Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with OP = 3. The point S is directly above the mid-point T of diagonal OQ such that TS = 3. Then

(A) the acute angle between OQ and OS is $\frac{\pi}{3}$

(B) the equation of the plane containing the triangle OQS is $x - y = 0$

(C) the length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$

(D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

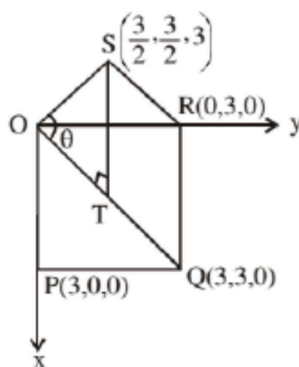
Ans. B, C, D

Sol. Points O, P, Q, R, S are (0, 0, 0), (3, 0, 0), (3, 3, 0), (0, 3, 0), $(\frac{3}{2}, \frac{3}{2}, 3)$ respectively

Angle between OQ and OS is $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

Equation of plane containing the points O, Q and S is $x - y = 0$

Perpendicular distance from P(3, 0, 0) to the plane



$$x - y = 0 \text{ is } \left| \frac{3-0}{\sqrt{2}} \right| = \frac{3}{\sqrt{2}}$$

Perpendicular distance from O(0, 0, 0) to the line RS:

$$\frac{x}{1} = \frac{y-3}{-1} = \frac{z}{2} \text{ is } \sqrt{\frac{15}{2}}$$

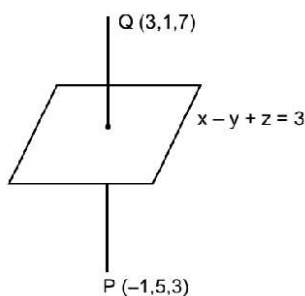
Q23. Let P be the image of the point (3, 1, 7) with respect to the plane $x - y + z = 3$. Then the equation of the plane passing through P and containing the straight line

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{1} \text{ is}$$

- (A) $x + y - 3z = 0$ (B) $3x + z = 0$ (C) $x - 4y + 7z = 0$ (D) $2x - y = 0$

Ans. C

Sol.



Get image of P(1, 3, 7) with respect to given plane i.e. Q(-1, 5, 3)

$$\begin{aligned} \text{Normal vector} &= \vec{OQ} \times (\hat{i} + 2\hat{j} + \hat{k}) \\ &= \hat{i} - 4\hat{j} + 7\hat{k} \end{aligned}$$

Equation of plane will be $x - 4y + 7z = \lambda$

Point (0, 0, 0) will lie in the plane

$$\Rightarrow 0 = \lambda$$

So the required plane is $x - 4y + 7z = 0$

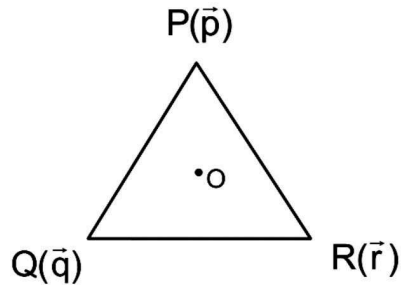
Q24. Let O be the origin and let PQR be an arbitrary triangle. Then point S is such that :

$$\vec{OP} \cdot \vec{OQ} + \vec{OR} \cdot \vec{OS} = \vec{OR} \cdot \vec{OP} + \vec{OQ} \cdot \vec{OS} = \vec{OQ} \cdot \vec{OR} + \vec{OP} \cdot \vec{OS}$$

(A) incentre (B) circumcentre (C) orthocentre (D) centroid

Ans. C

Sol.



$$\vec{p} \cdot \vec{q} + \vec{r} \cdot \vec{s} = \vec{r} \cdot \vec{p} + \vec{q} \cdot \vec{s} = \vec{q} \cdot \vec{r} + \vec{p} \cdot \vec{s}$$

$$\vec{p} \cdot (\vec{q} - \vec{r}) - \vec{s}(\vec{q} - \vec{r}) = 0 \Rightarrow \vec{PS} \cdot \vec{QR} = 0$$

$$\text{Similarly } \vec{PQ} \cdot \vec{SR} = 0$$

S is orthocentre of the triangle

Paragraph

Let O be the origin, and $\vec{OX}, \vec{OY}, \vec{OZ}$ be three unit vectors in the directions of the sides $\vec{QR}, \vec{RP}, \vec{PQ}$ respectively, of a triangle PQR.

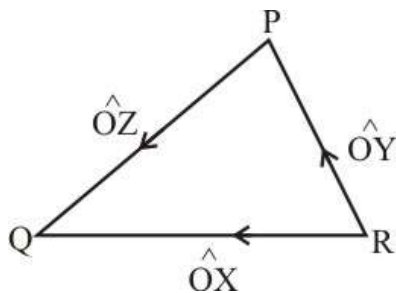
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Q25. $|\vec{OX} \times \vec{OY}|$

(A) $\sin(P + R)$ (B) $\sin 2R$ (C) $\sin(P + Q)$ (D) $\sin(Q + R)$

Ans. C

Sol.



$$\cos R = - \hat{OX} \cdot \hat{OY}$$

$$\Rightarrow |\cos R| = |\vec{OX} \cdot \vec{OY}|$$

$$|\vec{OX} \cdot \vec{OY}| = |\sin R| = \sin(\pi - (P + Q)) = |\sin(P + Q)| = \sin(P + Q)$$

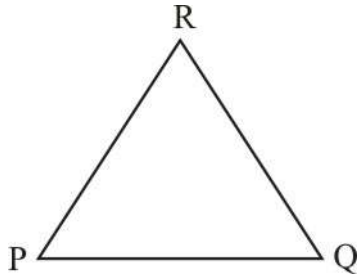
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Q26. If the triangle PQR varies, then the minimum value of $\cos(P + Q) + \cos(Q + R) + \cos(R + P)$ is :

- (A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{5}{3}$ (D) $\frac{5}{3}$

Ans. A

Sol.



$$\cos(P + Q) + \cos(Q + R) + \cos(R + P) = -\cos R - \cos P - \cos Q$$

$$\text{In any } \triangle, \max \text{ of } \cos P + \cos Q + \cos R = \frac{3}{2}$$

$$\text{So minimum value of the give expresstion is } -\frac{3}{2}$$

Q27. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is :

- (A) $-14x + 2y + 15z = 3$ (B) $14x - 2y + 15z = 27$ (C) $14x + 2y - 15z = 1$
(D) $14x + 2y + 15z = 31$

Ans. D

Sol. Let plane be

$$a(x - 1) + b(y - 1) + c(z - 1) = 0$$

$$\text{Now, direction ratio of its normal} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 3 & -6 & -2 \end{bmatrix} = \hat{i}(-14) - \hat{j}(2) + \hat{k}(-15)$$

$$\text{So, } -14(x - 1) - 2(y - 1) - 15(z - 1) = 0$$

$$14x + 2y + 15z = 31$$

Q28. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a} \cdot \vec{b} = 0$. For some $x, y \in \mathbb{R}$, let $\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b})$. If $|\vec{c}| = 2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2 \alpha$ is _____.

Ans. 3

Sol. $\vec{c} = x\vec{a} + y\vec{b} + \vec{a} \times \vec{b}$ & $\vec{a} \cdot \vec{b} = 0$

$$\vec{a} \cdot \vec{c} = \vec{b} \cdot \vec{c} = \alpha$$

$$\vec{c} \cdot \vec{a} = \vec{c} \cdot \vec{b} = 2 \cos \alpha \quad x = y = 2 \cos \alpha$$

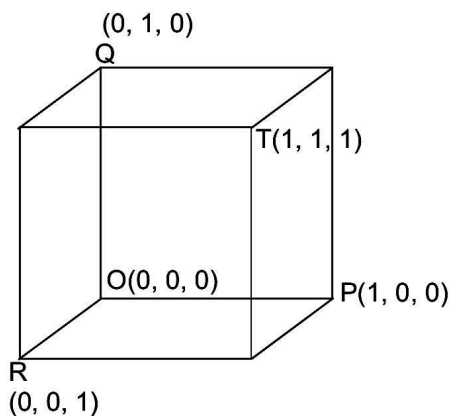
$$|\vec{c}|^2 = x^2 + y^2 + |\vec{a} \times \vec{b}|^2 = 2(4\cos^2 \alpha) + 1 - 0$$

$$4 = 8 \cos^2 \alpha + 1 \quad 8\cos^2 \alpha = 3$$

- Q29. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0,0,0) is the origin. Let $S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p} = \vec{SP}, \vec{q} = \vec{SQ}, \vec{r} = \vec{SR}$ and $\vec{t} = \vec{ST}$, then the value of $|(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})|$ is _____.

Ans. 0.5

Sol.



$$\text{point } S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \text{ point } T(1, 1, 1) \vec{p} = \vec{SP} = \frac{\hat{i} - \hat{j} - \hat{k}}{2} \vec{q} = \vec{SQ} = \frac{-\hat{i} - \hat{j} - \hat{k}}{2} \vec{r} = \vec{SR} = \frac{-\hat{i} - \hat{j} + \hat{k}}{2} \vec{t} = \vec{ST} =$$

$$\text{Now } \vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ -1 & 1 & -1 \end{vmatrix} \times \frac{1}{4} = \frac{1}{4}(2\hat{i} + \hat{j}) = \frac{\hat{i} + \hat{j}}{2} \vec{r} \times \vec{t} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} \times \frac{1}{4} = \frac{-2\hat{i} + 2\hat{j}}{4} = \frac{-\hat{i} + \hat{j}}{2}$$

$$\text{Now } (\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ -1 & 1 & 0 \end{vmatrix} \times \frac{1}{4} = \frac{\hat{k}}{2} \Rightarrow |(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})| = \frac{1}{2} = 0.5$$

- Q30. Let $P_1: 2x + y - z = 3$ and $P_2: x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE ?

- (A) The line of intersection of P_1 and P_2 has direction ratios 1, 2, -1
- (B) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2
- (C) The acute angle between P_1 and P_2 is 60°
- (D) If P_3 is the plane passing through the point (4, 2, -2) and perpendicular to the line of intersection of P_1 and P_2 , then the distance of the point (2, 1, 1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

Ans. C, D

Sol. Direction ratio of common line is $\vec{n}_1 \times \vec{n}_2$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{i}(3) - \hat{j}(3) + \hat{k}(3) = 3(\hat{i} - \hat{j} + \hat{k})$$

$$(B) \frac{x - 4/3}{3} = \frac{y - 1/3}{-3} = \frac{z}{3}$$

This is || to line of intersection

$$(C) \cos \theta = \frac{\vec{x}_1 \cdot \vec{x}_2}{|\vec{x}_1||\vec{x}_2|} = \frac{2 + 2 - 1}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2} \quad \theta = \frac{\pi}{3}$$

(D) $P_3 : x - y + z = 1$ satisfy $(4, 2, -2)$

$$4 - 2 - 2 = 1 \quad x - y + z = 4$$

$$\text{distance of the point } (2, 1, 1) \text{ from the plane } P_3 = \left| \frac{2 - 1 + 1 - 4}{\sqrt{3}} \right| \Rightarrow \frac{2}{\sqrt{3}}$$

Q31. Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$ lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____ .

Ans. 8

Sol. $Q(\alpha, \beta, \gamma)R(\alpha, \beta, -\gamma)Q \frac{x-\alpha}{1} = \frac{y-\beta}{1} = \frac{z-\gamma}{0} = \frac{-2(\alpha+\beta-3)}{2} x = 3 - \beta, y = 3 - \alpha, z = \gamma Q(3 - \beta), y$
 $\gamma^2 = 16 \Rightarrow \gamma = 4 P(3, 3, 4) \therefore PR = 8R(3, 3, -4)$

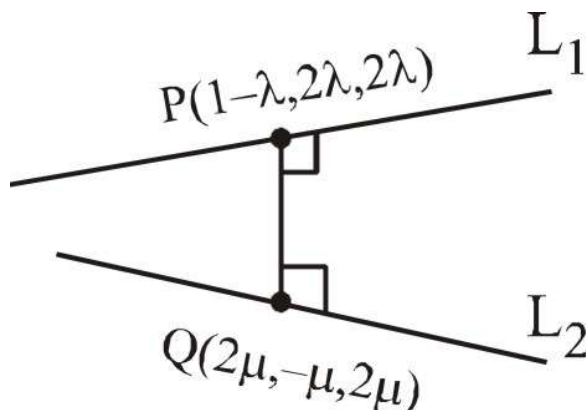
Q32. Let L_1 and L_2 denote the lines $\vec{r} = \hat{i} + \lambda(-\hat{i} + 2\hat{j} + 2\hat{k}), \lambda \in R, \vec{r} = \mu(2\hat{i} - \hat{j} + 2\hat{k}), \mu \in R$, respectively. If L_3 is line which is perpendicular to both L_1 and L_2 and cuts both of them then which of the following options describe (s) L_3

$$(A) \vec{r} = \frac{1}{3}(2\hat{i} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R \quad (B) \vec{r} = \frac{2}{9}(2\hat{i} - \hat{j} + 2\hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$$

$$(C) \vec{r} = \frac{2}{9}(4\hat{i} + \hat{j} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R \quad (D) \vec{r} = t(2\hat{i} + 2\hat{j} - \hat{k}), t \in R$$

Ans. A, B, C

Sol.



$$\overrightarrow{PQ} = (1 - \lambda - 2\mu, 2\lambda + \mu, 2\lambda - 2\mu)PQ \perp L_1 \quad \text{and}$$

$$PQ \perp L_2$$

$$\text{mid point of } PQ \left(\frac{2}{3}, 0, \frac{1}{3} \right)$$

Q33. Three lines

$$L_1 : \vec{r} = \lambda \hat{i}, \lambda \in R$$

$$L_2 : \vec{r} = \hat{k} + \mu \hat{j}, \mu \in R \quad \text{and,}$$

$L_3 : \vec{r} = \hat{i} + \hat{j} + v\hat{k}, v \in R$ are given. For which point (s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear

$$(A) \hat{k} + \hat{j} \quad (B) \hat{k} - \frac{1}{2}\hat{j} \quad (C) \hat{k} + \frac{1}{2}\hat{j} \quad (D) \hat{k}$$

Ans. B, C

$$\text{Sol. } P(\lambda, 0, 0)Q(0, \mu, 1)R(1, 1, v)P, Q, R \text{ are collinear } \overrightarrow{PQ} = t\overrightarrow{QR} \quad t \in R \quad \overrightarrow{PQ} = (-\lambda, \mu, 1)\overrightarrow{QR} = (-1,$$

Q34. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. Consider a vector

$$\vec{c} = \alpha\vec{a} + \beta\vec{b}, \alpha, \beta \in R. \text{ If the}$$

projection of $\vec{c}$ on the vector $(\vec{a} + \vec{b})$ is $3\sqrt{2}$ then the minimum value of $(\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c}$ equals _____

Ans. 18

$$\text{Sol. } |\vec{a}|^2 = |\vec{b}|^2 = 6 \quad \hat{a} + \hat{b} = 3\hat{i} + 3\hat{j}$$

$$\vec{a} \cdot \vec{b} = 2 + 2 - 1 = 3$$

Projection of $\vec{c}$ on $\vec{a} + \vec{b}$ is

$$\frac{\vec{c} \cdot (\vec{a} + \vec{b})}{|\vec{a} + \vec{b}|} = 3\sqrt{2}$$

$$\frac{(\alpha\vec{a} + \beta\vec{b}) \cdot (\vec{a} + \vec{b})}{3\sqrt{2}} = 3\sqrt{2}$$

$$9(\alpha + \beta) = 18$$

$$\alpha + \beta = 2$$

$$\text{Let } y = (\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c}$$

$$= \vec{c} \cdot \vec{c} - (\alpha\vec{a} + \beta\vec{b}) \cdot (\vec{a} \times \vec{b})$$

$$= |\vec{c}|^2 - 0$$

$$\begin{aligned}
&= \alpha^2 |\vec{a}|^2 + \beta^2 |\vec{b}|^2 + 2\alpha\beta \vec{a} \cdot \vec{b} \\
&= 6\alpha^2 + 6\beta^2 + 6\alpha\beta \\
&= 6(\alpha^2 + \beta^2 + \alpha\beta) \quad (\alpha = 2 - \beta) \\
&= 6\left((2 - \beta)^2 + \beta^2 + (2 - \beta)\beta\right) \\
&= 6(4 - 4\beta + \beta^2 + \beta^2 + 2\beta - \beta^2) \\
&= 6(\beta^2 - 2\beta + 1 + 3) \\
y &= 6\left((\beta - 1)^2 + 3\right) \\
y_{\min} &= 6(0 + 3) = 18
\end{aligned}$$

Q35. Three lines are given by

$$\vec{r} = \lambda \hat{i}, \lambda \in R, \vec{r} = \mu(\hat{i} + \hat{j}), \mu \in R \text{ and } \vec{r} = v(\hat{i} + \hat{j} + \hat{k}), v \in R$$

Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the area of the triangle ABC is Δ then the value of $(6\Delta)^2$ equals

Ans. 0.75

Sol.

Let $A(\lambda, 0, 0)$, $B(\mu, \mu, 0)$, $C(v, v, v)$ lie on the plane $x + y + z = 1$ so $\lambda = 1, \mu = \frac{1}{2}, v = \frac{1}{3}$

$$\begin{aligned}
AB \times AC &= \begin{vmatrix} i & j & k \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 2 & 1 & 1 \end{vmatrix} = \frac{1}{6} \begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -2 & 1 & 1 \end{vmatrix} \\
&= \frac{1}{6} (i(1 - 0) - j(-1 - 0) + k(-1 + 2)) = \frac{1}{6} (i + j + k) \\
&= \frac{1}{6} (\hat{i} + \hat{j} + \hat{k}) \Delta AB = \frac{1}{2} |AB \times AC| \Delta = \frac{1}{2} \cdot \frac{1}{6}
\end{aligned}$$

Q36. In a triangle PQR, let $\vec{a} = \vec{QR}$, $\vec{b} = \vec{RP}$ and $\vec{c} = \vec{PQ}$. If

$$|\vec{a}| = 3, |\vec{b}| = 4, \text{ and } \frac{\vec{a} \cdot (\vec{c} \times \vec{b})}{\vec{c} \cdot (\vec{a} - \vec{b})} = \frac{|\vec{a}|}{|\vec{a}| + |\vec{b}|} \text{ then the value of } |\vec{a} \times \vec{b}|^2 \text{ is } \underline{\hspace{2cm}}$$

Ans. bonus

$$\begin{aligned}
\text{Sol. } \vec{a} &= \vec{QR}, \vec{b} = \vec{RP}, \vec{c} = \vec{PQ} \\
\Rightarrow \vec{a} + \vec{b} + \vec{c} &= \vec{0}
\end{aligned}$$

$$\Rightarrow \vec{c} = -\vec{a} - \vec{b}$$

Replacing in the given equation,

$$\frac{\vec{a} \cdot (-\vec{a} - \vec{b} - \vec{b})}{(-\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})} = \frac{|\vec{a}|}{|\vec{a}| + |\vec{b}|} = \frac{3}{3+4}$$

$$\Rightarrow \frac{|\vec{a}|^2 + 2\vec{a} \cdot \vec{b}}{|\vec{a}|^2 - |\vec{b}|^2} = \frac{3}{7}$$

$$\Rightarrow \frac{9 + 2\vec{a} \cdot \vec{b}}{9 - 16} = \frac{3}{7}$$

Now from Lagrange's Identity

$$= 9 \times 16 - (-6)^2$$

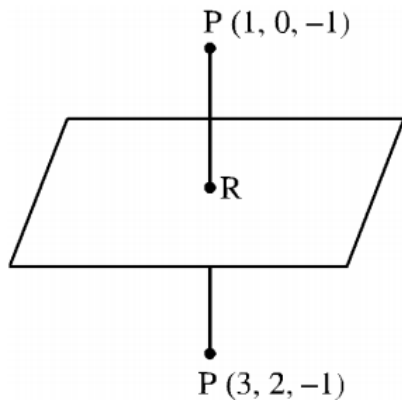
$$= 144 - 36 = 108$$

Q37. Let $\alpha, \beta, \gamma, \delta$ be real numbers such that $\alpha^2 + \beta^2 + \gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point $(3, 2, -1)$ is the mirror image of the point $(1, 0, -1)$ with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE?

- (A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$ (C) $\delta - \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

Ans. A, B, C

Sol.



R is mid point of PQ

$$\therefore R(2, 1, -1)$$

and it lies on plane equation of plane is $\alpha a + \beta y + \gamma z = \delta$

$$\therefore 2\alpha + \beta - \gamma = \delta \quad \dots(1)$$

Normal vector to plane is

$$\vec{n} = 2\mathbf{i} + 2\mathbf{j}$$

$$\therefore \frac{\alpha}{2} = \frac{\beta}{2} = \frac{\gamma}{0} = k$$

$$\therefore \alpha = 2k, \beta = 2k, \gamma = 0 \quad \dots(2)$$

$$\text{and } \alpha + \gamma = 1 \text{ (given)} \quad \dots(3)$$

from (2) and (3)

$$\therefore \alpha = 1, \beta = 1, \gamma = 0$$

and from (1)

$$2(1) + 1 - 0 = \delta$$

$$\delta = 3$$

Now :

$$\alpha + \beta = 2$$

$$\delta - \gamma = 3$$

$$\delta + \beta = 4$$

So, A, B, C are correct

- Q38. Let a and b be positive real numbers. Suppose $\vec{PS} = a\hat{i} + b\hat{j}$ and $\vec{PQ} = a\hat{i} + b\hat{j}$ and are adjacent sides of a parallelogram PQRS. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w} = \hat{i} + \hat{j}$ along $\vec{PQ}$ and $\vec{PS}$, respectively. If $|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram PQRS is 8, then which of the following statements is/are TRUE?

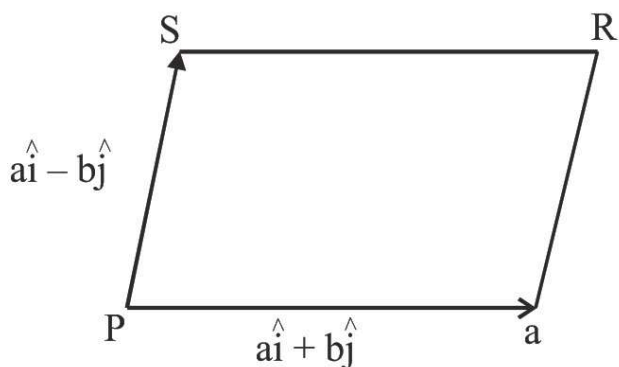
(A) $a + b = 4$ (B) $a - b = 2$

(C) The length of the diagonal PR of the parallelogram PQRS is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\vec{PQ}$ and $\vec{PS}$

Ans. A, C

Sol.



$$\vec{u} = \left(\frac{\vec{w} \cdot \vec{PQ}}{|\vec{PQ}|} \right) \hat{PQ} = \left(\frac{a - b}{\sqrt{a^2 + b^2}} \right) \left(\frac{a\hat{i} - b\hat{j}}{\sqrt{a^2 + b^2}} \right)$$

$$\vec{v} = \left(\frac{\vec{w} \cdot \vec{PS}}{|\vec{PS}|} \right) \hat{PS} = \left(\frac{a - b}{\sqrt{a^2 + b^2}} \right) \left(\frac{a\hat{i} - b\hat{j}}{\sqrt{a^2 + b^2}} \right)$$

$$|\vec{u}| + |\vec{v}| = |\vec{w}|$$

$$\frac{a + b}{\sqrt{a^2 + b^2}} = \sqrt{2}$$

Case - I

$$a > b$$

$$2a = \sqrt{2}\sqrt{a^2 + b^2}$$

$$2a^2 = a^2 + b^2$$

$$a^2 = b^2$$

$$a = b$$

Case - II

$$a < b$$

$$2a = \sqrt{2}\sqrt{a^2 + b^2}$$

$$a = b$$

$$\text{Area of the parallelogram} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & -b & 0 \\ a & b & 0 \end{vmatrix} = 8$$

$$|\hat{k} \cdot 2ab| = 8$$

$$ab = 4 \text{ and } a = b$$

$$a = b = 2$$

$$(A) a + b = 4$$

$$(C) \overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = 2a\hat{i}$$

$$|\overrightarrow{PR}| = 4$$

$$(D) \overrightarrow{PQ} + \overrightarrow{PS} = 2a\hat{i} + \lambda\vec{\omega}$$

$$\overrightarrow{PQ} - \overrightarrow{PS} = 2b\hat{j} \neq \lambda\vec{\omega}$$

Q39. Let L_1 and L_2 be the following straight lines.

$$L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3} \text{ and } L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$$

Suppose the straight line

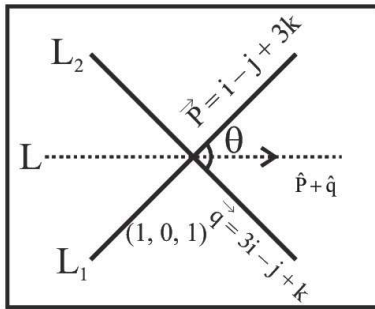
$$L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$$

lies in the plane containing L_1 and L_2 , and passes through the point of intersection of L_1 and L_2 . If the line L bisects the acute angle between the lines L_1 and L_2 , then which of the following statements is/are TRUE?

$$(A) \alpha - \gamma = 3 \quad (B) l + m = 2 \quad (C) \alpha - \gamma = 1 \quad (D) l + m = 0$$

Ans. A, B

Sol.



$$\vec{p} \cdot \vec{q} = 1 > 0 \Rightarrow \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\hat{p} + \hat{q} = \frac{1}{\sqrt{11}}(-2i - 2j + 4k)$$

DR's of L = -2, -2, 4

OR

$$L: \frac{x - \alpha}{l} = \frac{y - 1}{m} = \frac{z - \gamma}{-2}$$

$$l = 1$$

$$m = 1$$

Passes through (1, 0, 1)

$$\alpha = 2$$

$$g = -1$$

$$\alpha - g = 3 \text{ (A)}$$

$$l + m = 2 \text{ (B)}$$

Q40. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u} \cdot \vec{w} = 1, \vec{v} \cdot \vec{w} = 1, \vec{w} \cdot \vec{w} = 4$. If the volume of the parallelepiped, whose adjacent sides are represented by the $\vec{u}, \vec{v}$ vectors $\vec{w}$ and $\sqrt{2}$, is 7, then the value of $|\vec{3u} + \vec{5v}|$ is ____.

Ans. 7

Sol. $\vec{u} \cdot \vec{v} \neq 0$

$$\vec{u} \cdot \vec{w} = 1, \vec{v} \cdot \vec{w} = 1, \vec{w} \cdot \vec{w} = 4, [\vec{u}\vec{v}\vec{w}] = \sqrt{2}$$

$$|\vec{3u} + \vec{5v}| = ?, \vec{u} \& \vec{v} \text{ are unit vectors}$$

$$[\vec{u}\vec{v}\vec{w}] = \sqrt{2}$$

$$\Rightarrow [\vec{u}\vec{v}\vec{w}]^2 = 2$$

$$\Rightarrow \begin{vmatrix} \vec{u} \cdot \vec{u} & \vec{u} \cdot \vec{v} & \vec{u} \cdot \vec{w} \\ \vec{v} \cdot \vec{u} & \vec{v} \cdot \vec{v} & \vec{v} \cdot \vec{w} \\ \vec{w} \cdot \vec{u} & \vec{w} \cdot \vec{v} & \vec{w} \cdot \vec{w} \end{vmatrix} = 2$$

$$\Rightarrow \begin{vmatrix} 1 & x & 1 \\ x & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} = 2 \text{ (Let } \vec{u} \cdot \vec{v} = x)$$

$$\begin{aligned}
&\Rightarrow 1(4-1) - x(4x-1) + 1(x-1) = 2 \\
&\Rightarrow 3 - 4x^2 + x + x - 1 = 2 \\
&\Rightarrow 4x^2 - 2x = 0 \\
&\Rightarrow x = \left\{0, \frac{1}{2}\right\} \Rightarrow \vec{u} \cdot \vec{v} = 0 \text{ (rejected) } \text{ or } \vec{u} \cdot \vec{v} = \frac{1}{2} \\
&\text{Now, } |3\vec{u} + 5\vec{v}|^2 = 9\vec{u} \cdot \vec{u} + 25\vec{v} \cdot \vec{v} + 30\vec{u} \cdot \vec{v} \\
&9 + 25 + 30 \times \frac{1}{2} \\
&= 49 \\
&\Rightarrow |3\vec{u} + 5\vec{v}| = 7
\end{aligned}$$

- Q41. Let O be the origin and $\vec{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{OB} = \hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{OC} = \frac{1}{2}(\vec{OB} - \lambda\vec{OA})$ for some $\lambda > 0$. If $|\vec{OB} \times \vec{OC}| = \frac{9}{2}$, then which of the following statements is (are) TRUE ?
- (A) Projection of $\vec{OC}$ on $\vec{OA}$ is $-\frac{3}{2}$ (B) Area of the triangle OAB is $\frac{9}{2}$
- (C) Area of the triangle ABC is $\frac{9}{2}$
- (D) The acute angle between the diagonals of the parallelogram with adjacent sides $\vec{OA}$ and $\vec{OC}$ is $\frac{\pi}{3}$

Ans. A, B, C

Sol. $\vec{OA} = \vec{a} = 2\vec{i} + 2\vec{j} + \vec{k}$
 $\vec{OB} = \vec{b} = \vec{i} - 2\vec{j} + 2\vec{k}$
 $\vec{OC} = \vec{c} = \frac{1}{2}(\vec{b} - \lambda\vec{a})$
 $|\vec{OB} \times \vec{OC}| = \frac{9}{2}$
 $\left| \vec{b} \times \frac{(\vec{b} - \lambda\vec{a})}{2} \right| = \frac{9}{2}$
 $\frac{|\lambda|}{2} |\vec{b} \times \vec{a}| = \frac{9}{2}$
 $\frac{|\lambda|}{2} \cdot 9 = \frac{9}{2} \Rightarrow \lambda \pm 1 \ (\lambda > 0)$
 $\lambda = 1$
 $\vec{c} = \frac{\vec{b} - \vec{a}}{2}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = 6\vec{i} - 3\vec{j} - 6\vec{k}$$

$$\vec{a} \times \vec{b} = 3(2\vec{i} - \vec{j} - 2\vec{k})$$

(A) Projection of $\vec{OC}$ on $\vec{OA}$

$$\frac{\vec{c} \cdot \vec{a}}{|\vec{a}|} = \frac{(\vec{b} - \vec{a}) \cdot \vec{a}}{2|\vec{a}|} = \frac{0 - |\vec{a}|^2}{2|\vec{a}|} = \frac{-3}{2}$$

$$(B) \text{ Area of } \Delta OAB = \frac{1}{2} |\vec{a} \times \vec{b}| = \frac{9}{2}$$

$$(C) \text{ Area of } \Delta ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

$$= \frac{1}{2} \left| \vec{a} \times \vec{b} + \vec{b} \times \left(\frac{\vec{b} - \vec{a}}{2} \right) + \left(\frac{\vec{b} - \vec{a}}{2} \right) \times \vec{a} \right|$$

$$= \frac{1}{2} |\vec{a} \times \vec{b}| = \frac{9}{2}$$

$$(D) \vec{d}_1 = \frac{\vec{a} + \vec{c}}{2} = \frac{\vec{a} + \left(\frac{\vec{b} - \vec{a}}{2} \right)}{2} = \frac{\vec{a} + \vec{b}}{4} = \frac{3i + 3k}{4}$$

$$\vec{d}_2 = \frac{\vec{a} - \vec{c}}{2} = \frac{\vec{a} - \left(\frac{\vec{b} - \vec{a}}{2} \right)}{2} = \frac{3\vec{a} - \vec{b}}{4} = \frac{5i + 8j + k}{4}$$

$$\cos \theta = \frac{|\vec{d}_1 \cdot \vec{d}_2|}{|\vec{d}_1| |\vec{d}_2|}$$

$$\cos \theta = \frac{1}{\sqrt{5}}$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

As (ABC)

Q42. Let α, β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point $(0, 1, 0)$ from the plane P. The value of $|M|$ is_____.

Ans. 0.95, 1.05

Sol. $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$

$$\Rightarrow \Delta_x = 0$$

$$\Rightarrow \begin{vmatrix} \alpha & 2 & 4 \\ \beta & 1 & 0 \\ \gamma - 1 & 8 & 9 \end{vmatrix} = 0$$

$$|M| = \begin{vmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

$$|M| = \alpha - 2\beta + \gamma = 1$$

Q43. Let α, β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point $(0, 1, 0)$ from the plane P. The value of D is _____.

Ans. 1.45, 1.55

Sol. Plane P $x - 2y + z = 1$

$$\sqrt{D} = \left| \frac{0 - 2 + 0 - 1}{\sqrt{1 + 4 + 1}} \right| = \frac{3}{\sqrt{6}}$$

$$D = \frac{9}{6} = 1.5$$

Q44. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes.

$$\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$$

$$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in R$$

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in R$$

be three vectors such that $b_2b_3 > 0$, $\vec{a} \cdot \vec{b} = 0$ and $\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}$.

Then, which of the following is /are TRUE.

$$(A) \vec{a} \cdot \vec{c} = 0 \quad (B) \vec{b} \cdot \vec{c} = 0 \quad (C) |\vec{b}| > \sqrt{10} \quad (D) |\vec{c}| \leq \sqrt{11}$$

Ans. B, C, D

$$\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$$

$$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$$

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}$$

multiply & compare

$$b_2c_3 - b_3c_2 = c_1 - 3 \quad \dots\dots\dots(1)$$

$$c_3 - b_3c_1 = 1 - c_2 \quad \dots\dots\dots(2)$$

$$c_2 - b_2c_1 = 1 + c_3 \quad \dots\dots\dots(3)$$

$$(1)\hat{i} - (2)\hat{j} + (3)\hat{k}$$

$$\hat{i}(b_2c_3 - b_3c_2) - \hat{j}(c_3 - b_3c_1) + \hat{k}(c_2 - b_2c_1)$$

$$= c_1\hat{i} + c_2\hat{j} + c_3\hat{k} - 3\hat{i} - \hat{j} + \hat{k}$$

$$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$$

Take dot product with $\vec{b}$

$$0 = \vec{c} \cdot \vec{b} - \vec{a} \cdot \vec{b}$$

$$\vec{b} \cdot \vec{c} = 0$$

$$\vec{b} \perp \vec{c}$$

$$\vec{b} \wedge \vec{c} = 90^\circ$$

Take dot product with $\vec{c}$

$$0 = |\vec{c}|^2 - \vec{a} \cdot \vec{c}$$

$$\vec{a} \cdot \vec{c} = |\vec{c}|^2$$

$$\vec{a} \cdot \vec{c} \neq 0$$

$$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$$

Squaring

$$|\vec{b}|^2 |\vec{c}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a}$$

$$|\vec{b}|^2 |\vec{c}|^2 = |\vec{c}|^2 + 11 - 2|\vec{c}|^2$$

$$|\vec{b}|^2 |\vec{c}|^2 = 11 - |\vec{c}|^2$$

$$|\vec{c}|^2 = \frac{11}{|\vec{b}|^2 + 1}$$

$$|\vec{c}| \leq \sqrt{11}$$

$$\text{given } \vec{a} \cdot \vec{b} = 0$$

$$b_2 - b_3 = -3 \quad \text{also}$$

$$b_2^2 + b_3^2 - 2b_2b_3 = 9 \quad b_2b_3 > 0$$

$$b_2^2 + b_3^2 = 9 + 2b_2b_3$$

$$b_2^2 + b_3^2 = 9 + 2b_2b_3 > 9$$

$$b_2^2 + b_3^2 > 9$$

$$|\vec{b}| = \sqrt{1 + b_2^2 + b_3^2}$$

$$|\vec{b}| > \sqrt{10}$$

Q45. Let P_1 and P_2 be two planes given by

$$P_1 : 10x + 15y + 12z - 60 = 0,$$

$$P_2 : -2x + 5y + 4z - 20 = 0,$$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ?

$$(A) \frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5} \quad (B) \frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3} \quad (C) \frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$$

$$(D) \frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$$

Ans. A, B, D

Sol. Edge of Tetrahedron can be (i) common intersection line

(ii) Line totally in P_2/P_1 and meeting intersecting line.

(iii) Line intersecting both P_1 and P_2

(i) Common intersecting line is parallel to $\vec{n}_1 \times \vec{n}_2$

$$10a + 15b + 12c = 0$$

$$-2a + 5b + 4c = 0$$

i.e. it has direction ratio 0, -4, 5 and a point on it can be (0, 0, 5)

$$\frac{x}{0} = \frac{y}{-4} = \frac{z-5}{5} \text{ Not possible}$$

(ii) Line in P_2/P_1 must be perpendicular to their normals. option 'D' Line in plane P_2 and it should also cut common line

$\Rightarrow$ 0, -4 λ , 5 λ , + 5 should lie on D.

$$\frac{0}{1} = \frac{-4\lambda - 4}{-2} = \frac{5\lambda + 5}{3} \quad \text{which is true for } \lambda = -1.$$

Hence 'D' is correct.

(iii) In option 'C' is parallel to normal at 'P₂' but it is passing through (0, 4, 0) which is a point on intersecting lines. Hence, this can't be an edge of tetrahedron (A) and (B) cuts both P_1 and P_2 and do not passes through common intersecting lines hence are possible.

(A) and (B) are possible, Hence A, B, D are correct

Q46. Let S be the reflection of a point Q with respect to the plane given by

$\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ where t, p are real parameters and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ?

$$(A) 3(\alpha + \beta) = -101 \quad (B) 3(\alpha + \gamma) = -71 \quad (C) 3(\gamma + \alpha) = -86 \quad (D) 3(\alpha + \beta + \gamma) = -121$$

Ans. A, B, C

Sol. $\vec{r} = \hat{k} + (-\hat{i} + \hat{k})p + t(-\hat{i} + \hat{j})$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} - \hat{k}$$

Equation of Plane $x + y + z = 1$

$$Q = (10, 15, 20)$$

$$F = (10 + d, 15 + d, 20 + d)$$

$$10 + d + 10 + d + 20 + d = 1$$

$$3d = -44$$

$$\frac{\alpha + 10}{2} = 10 + d$$

$$3\alpha = 30 - 2(44) = -58$$

$$3\beta = 45 - 88 = -43$$

$$3\gamma = 60 - 88 = -28$$

Q47. Let l_1 and l_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$, respectively. Let X set of all the planes H that contain the line l_1 . For a plane H, let d(H) denote the smaller

possible distance between the points of l_2 and H. Let H_0 be a plane in X for which d(H) maximum value of d(H) as H varies over all planes in X.

Match each entry in List-I to the correct entries in List-II.

List - I	I
(P) The value of $d(H_0)$ is	(1)
(Q) The distance of the point (0,1,2) from H_0 is	(2)
(R) The distance of origin from H_0 is	(3)
(S) The distance of origin from the point of intersection of planes $y = z$, $x = 1$ and H_0 is	(4)
	(5)

(A) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (5); (S) $\rightarrow$ (1)

(B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (3); (S) $\rightarrow$ (1)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (3); (S) $\rightarrow$ (2)

(D) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (2)

Ans. 650ad68207534a8a63bfbf45

Sol. $\vec{r}_1 = \lambda(i + j + k); \quad \vec{r}_2 = (j - k) + \mu(i + k)$

H_0 will be plane containing l_1 & parallel to l_2 .

$\therefore$ $\perp$ r vector of plane parallel to lines l_1 & l_2 is $\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$

$$\Rightarrow i[1] - j[1 - 1] + k[-1]$$

$$\Rightarrow i - k$$

$\therefore$ Plane will be $x + 0 \cdot y - z = \lambda$; passes through origin

$$\therefore \lambda = 0$$

$\therefore$ Plane $\Rightarrow x - z = 0 \rightarrow H_0$ Plane

(P) $d(H_0) = 1$ distance of point (0, 1, -1) from H

$$d = \frac{|0 - (-1)|}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad P \rightarrow 5$$

$$(Q) d = \left| \frac{0 - 2}{\sqrt{2}} \right| = \sqrt{2} \quad Q \rightarrow 4$$

$$(R) d = \left| \frac{0}{\sqrt{2}} \right| = 0 \quad R \rightarrow 3$$

(S) Point of intersection will be (1,1,1)

$$d = \sqrt{1 + 1 + 1} = \sqrt{3}$$

(B)

- Q48. Let the position vector of the points P, Q, R and S be $\vec{a} = \hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{b} = 3\hat{i} + 6\hat{j} + 3\hat{k}$, $\vec{c} = \frac{17}{5}\hat{i} + \frac{16}{5}\hat{j} + 7\hat{k}$ and $\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$ respectively. Then which of the following statements is true?
- (A) The points P, Q, R and S are NOT coplanar
- (B) $\frac{\vec{b} + 2\vec{d}}{3}$ is the position vector of a point which divides PR internally in the ratio 5 : 4
- (C) $\frac{\vec{b} + 2\vec{d}}{3}$ is the position vector of a point which divides PR externally in the ratio 5 : 4
- (D) The square of the magnitude of the vector $\vec{b} \times \vec{d}$ is 95

Ans. B

Sol. P(1, 2, -5), Q(3, 6, 3), R $\left(\frac{17}{5}, \frac{16}{5}, 7\right)$, S(2, 1, 1)

$$\frac{\vec{b} + 2\vec{d}}{3} = \frac{7\hat{i} + 8\hat{j} + 5\hat{k}}{3}$$

$$\Rightarrow \frac{17\lambda}{5} + 1 = \frac{7}{3}(\lambda + 1)$$

$$\Rightarrow 51\lambda + 15 = 35\lambda + 35$$

$$\Rightarrow 16\lambda = 20 \quad \Rightarrow \lambda = \frac{5}{4}$$

- Q49. Let $\mathbb{R}^3$ denote the three-dimensional space. Take two points P = (1,2,3) and Q = (4,2,7). Let dist(X, Y) denote the distance between two points X and Y in $\mathbb{R}^3$. Let $S = \{X \in \mathbb{R}^3 : (\text{dist}(X, P))^2 - (\text{dist}(X, Q))^2 = 50\}$ and $T = \{Y \in \mathbb{R}^3 : (\text{dist}(Y, Q))^2 - (\text{dist}(Y, P))^2 = 50\}$. Then which of the following statements is(are) TRUE ?
- (A) There is a triangle whose area is 1 and all of whose vertices are from S
- (B) There are two distinct points L and M in T such that each point on the line segment LM is also in T
- (C) There are infinitely many rectangles of perimeter 48, two of whose vertices are from S and the other two vertices are from T
- (D) There is a square of perimeter 48, two of whose vertices are from S and the other two vertices are from T

Ans. A, B, C, D

Sol. $P(1,2,3)$
 $Q(4,2,7)$
 $S : 6x + 8z = 105$
 $T : 6x + 8z = 5$
 S & T represent two parallel plane
 Now check the options

Q50. A straight line drawn from the point $P(1, 3, 2)$, parallel to the line $\frac{x-2}{1} = \frac{y-4}{2} = \frac{z-6}{1}$, intersects the plane $L_1 : x - y + 3z = 6$ at the point Q . Another straight line which passes through Q and is perpendicular to the plane L_1 intersects the plane $L_2 : 2x - y + z = -4$ at the point R . Then which of the following statements is (are) TRUE?

- (A) The length of the line segment PQ is $\sqrt{6}$ (B) The coordinates of R are $(1, 6, 3)$
 (C) The centroid of the triangle PQR is $\left(\frac{4}{3}, \frac{14}{3}, \frac{5}{3}\right)$
 (D) The perimeter of the triangle PQR is $\sqrt{2} + \sqrt{6} + \sqrt{11}$

Ans. A, C

Sol. $P(1,3,2)$

$$\vec{r} = (1, 3, 2) + \lambda(1, 2, 1)$$

Line intersects plane L_1 at Q

$$\text{Let } Q(1 + \lambda, 3 + 2\lambda, 2 + \lambda)$$

$$(1 + \lambda) - (3 + 2\lambda) + 3(2 + \lambda) = 6$$

$$\Rightarrow \lambda = 1$$

$$\Rightarrow Q(2, 5, 3)$$

equations of line through Q

$$\vec{r} = (2, 5, 3) + \mu(1, -1, 3)$$

$$\text{Let } R(2 + \mu, 5 - \mu, 3 + 3\mu)$$

R will lie on plane L_2

$$2(2 + \mu) - (5 - \mu) + 3 + 3\mu = -4$$

$$\Rightarrow \mu = -1$$

$$\Rightarrow R(1, 6, 0)$$

$$(A) PQ = \sqrt{6}$$

$$(B) R(1, 6, 0)$$

(C) Centroid of $\Delta PQR \equiv \left(\frac{4}{3}, \frac{14}{3}, \frac{5}{3}\right)$

(D) Perimeter of $\Delta PQR = \sqrt{6} + \sqrt{11} + \sqrt{13}$

- Q51. Let $\vec{OP} = \frac{\alpha-1}{\alpha}\hat{i} + \hat{j} + \hat{k}$, $\vec{OQ} = \hat{i} + \frac{\beta-1}{\beta}\hat{j} + \hat{k}$ and $\vec{OR} = \hat{i} + \hat{j} + \frac{1}{2}\hat{k}$ be three vectors, where $\alpha, \beta \in \mathbb{R} - \{0\}$ and O denotes the origin. If $(\vec{OP} \times \vec{OQ}) \cdot \vec{OR} = 0$ and the point $(\alpha, \beta, 2)$ lies on the plane $3x + 3y - z + l = 0$, then the value of l is _____.

Ans. 5

Sol.
$$\begin{vmatrix} \frac{\alpha-1}{\alpha} & 1 & 1 \\ 1 & \frac{\beta-1}{\beta} & 1 \\ 1 & 1 & \frac{1}{2} \end{vmatrix} = 0$$

$$\left(\frac{\alpha-1}{\alpha}\right) \left[\frac{\beta-1}{2\beta} - 1\right] + \frac{1}{2} + \left(1 - \frac{(\beta-1)}{\beta}\right) = 0$$

$$\frac{-(\alpha-1)(\beta+1)}{2\alpha\beta} + \frac{1}{2} \cdot \frac{\beta\alpha}{\alpha\beta} + \frac{1}{\beta} \cdot \frac{2\alpha}{2\alpha} = 0$$

$$-\alpha\beta - \alpha + \beta + 1 + \alpha\beta + 2\alpha = 0$$

$$\alpha + \beta + 1 = 0$$

and $3(a+b) - 2 + l = 0$

$$l = 5$$

- Q52. Let $\vec{p} = 2\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{q} = \hat{i} - \hat{j} + \hat{k}$. If for some real numbers α, β , and γ , we have $15\hat{i} + 10\hat{j} + 6\hat{k} = \alpha(2\vec{p} + \vec{q}) + \beta(\vec{p} - 2\vec{q}) + \gamma(\vec{p} \times \vec{q})$ then the value of γ is _____.

Ans. 2

Sol. $\vec{p} = 2\hat{i} + \hat{j} + 3\hat{k}$

$$\vec{q} = \hat{i} - \hat{j} + \hat{k}$$

$$15\hat{i} + 10\hat{j} + 6\hat{k} = \alpha(2\vec{p} + \vec{q}) + \beta(\vec{p} - 2\vec{q}) + \gamma(\vec{p} \times \vec{q})$$

Take dot with $\vec{p} \times \vec{q}$

$$(15\hat{i} + 10\hat{j} + 6\hat{k}) \cdot (\vec{p} \times \vec{q}) = \alpha(2\vec{p} + \vec{q}) \cdot (\vec{p} \times \vec{q}) + \beta(\vec{p} - 2\vec{q}) \cdot (\vec{p} \times \vec{q}) + \gamma(\vec{p} \times \vec{q}) \cdot (\vec{p} \times \vec{q})$$

$$= 0 + 0 + \gamma|\vec{p} \times \vec{q}|^2$$

Now, $\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 3 \\ 1 & -1 & 1 \end{vmatrix}$

$$= \hat{i}(4) - \hat{j}(-1) + \hat{k}(-3)$$

$$= 4\hat{i} + \hat{j} - 3\hat{k}$$

$$|\vec{p} \times \vec{q}| = \sqrt{16 + 1 + 9}$$

$$\Rightarrow \sqrt{26}$$

$$(15\hat{i} + 10\hat{j} + 6\hat{k}) (4\hat{i} + \hat{j} - 3\hat{k}) = \gamma 26$$

$$\frac{60 + 10 - 18}{26} = \gamma$$

$$\gamma = 2$$

- Q53. Let $\gamma \in \mathbb{R}$ be such that lines $L_1 : \frac{x+11}{1} = \frac{y+21}{2} = \frac{z+29}{3}$ and $L_2 : \frac{x+16}{3} = \frac{y+11}{2} = \frac{z+4}{\gamma}$ intersect. Let R_1 be the point of intersection of L_1 and L_2 . Let $O = (0, 0, 0)$, and $\hat{n}$ denote a unit normal vector to the plane containing both the lines L_1 and L_2 :

Match each entry in List-I to the correct entry in List -II.

List -I	List -II
(P) γ equals	(1) $-\hat{i} - \hat{j} + \hat{k}$
(Q) A possible choice for $\hat{n}$ is	(2) $\sqrt{\frac{3}{2}}$
(R) $\overrightarrow{OR_1}$ equals	(3) 1
(S) A possible value of $\overrightarrow{OR_1} \cdot \hat{n}$ is	(4) $\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$
	(5) $\sqrt{\frac{2}{3}}$

The correct option is :

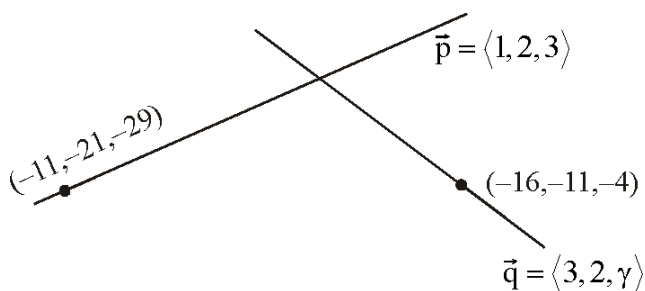
- (A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)
 (B) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (2)
 (C) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (4); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)
 (D) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

Ans. 666196783ac57f92d751f888

Sol. (P) Lines are intersecting.

Hence lines are coplanar.

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$



$$\begin{vmatrix} 5 & -10 & -25 \\ 1 & 2 & 3 \\ 3 & 2 & \gamma \end{vmatrix} = 0$$

$$\Rightarrow \gamma = 1 \quad \text{Hence (P)} \rightarrow (3)$$

(Q) normal

$$\vec{n} = \vec{p} \times \vec{q}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = -4\hat{i} + 8\hat{j} - 4\hat{k}$$

Hence a possible choice of $\hat{n}$ is $\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$, then (Q) $\rightarrow$ (4)

(R)

for POI of two lines

$$\langle \lambda - 11, 2\lambda - 21, 3\lambda - 29 \rangle$$

$$= \langle 3\mu - 16, 2\mu - 11, \mu - 4 \rangle$$

By comparing

$$\mu = 5$$

$$\lambda = 10$$

hence POI is $(-1, -1, 1)$.

$$\text{then } \overrightarrow{OR_1} = -\hat{i} - \hat{j} + \hat{k}$$

Hence (R) $\rightarrow$ (1)

$$(S) \overrightarrow{OR_1} \cdot \hat{n}$$

$$= (-\hat{i} - \hat{j} + \hat{k}) \cdot \left(\frac{\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{6}} \right)$$

$$= \sqrt{\frac{2}{3}}$$

Hence, (S) $\rightarrow$ (5)

Overall correct option is (C)

Exercise C1

Q1. Given $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) \cdot \hat{k} = \frac{\pi}{2}$, $\vec{a} \cdot \vec{c} = 4$, then

(A) $[\vec{a}\vec{b}\vec{c}]^2 = |\vec{a}|$ (B) $[\vec{a}\vec{b}\vec{c}] = |\vec{a}|$ (C) $[\vec{a}\vec{b}\vec{c}] = 0$ (D) $[\vec{a}\vec{b}\vec{c}] = |\vec{a}|^2$

Ans. D

Sol. $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$ $\vec{b} = \hat{i} - \hat{j} + \hat{k}$

$$\vec{a} \cdot \vec{b} = 0$$

$$x - y + 2 = 0$$

$$x - y = -2 \quad \dots\dots\dots(1)$$

$$\vec{a} \cdot \vec{c} = 4$$

$$x + 2y = 4 \quad \dots\dots\dots(2)$$

from (1) & (2)

$$x = 0, y = 2$$

$$[\vec{a} \cdot \vec{b} \cdot \vec{c}] = \begin{vmatrix} 1 & 2 & 2 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix}$$

$$= -2(-1) + 2(2 + 1)$$

$$= +2 + 6$$

$$= 8$$

$$|\vec{a}|^2 = 0 + 4 + 4 = 8$$

Q2. For a non zero vector $\vec{A}$ if the equations $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C}$ and $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$ hold simultaneously, then:

(A) $\vec{A}$ is perpendicular to $\vec{A} - \vec{C}$ (B) $\vec{A} = \vec{B}$ (C) $\vec{B} = \vec{C}$ (D) $\vec{C} = \vec{A}$

Ans. C

Sol. $\vec{A} \cdot \vec{B} = 0$

$$\vec{A} \cdot (\vec{B} - \vec{C}) = 0 \quad \dots\dots\dots(1)$$

from (1) & (2) either $\vec{A} = 0$

$$\vec{A} \times \vec{A} \times \vec{C} = 0$$

$$\vec{A} \times (\vec{B} - \vec{C}) = 0$$

$$\text{or } \vec{B} - \vec{C} = 0$$

$$\vec{B} = \vec{C}$$

Q3. The volume of the parallelopiped constructed on the diagonals of the faces of the given rectangular parallelopiped is m times the volume of the given parallelopiped. Then m is equal to:

- (A) 2 (B) 3 (C) 4 (D) None

Ans. A

Sol. Let sides of m parallelo piped are $\vec{a}, \vec{b}, \vec{c}$

then side of new $(\vec{a} + \vec{a}), (\vec{b} + \vec{c}), (\vec{c} + \vec{a})$

$$V = [\vec{a}\vec{b}\vec{c}]$$

$$V^1 = [\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}] = 2[\vec{a}\vec{b}\vec{c}]$$

$$= 2V$$

$$m = 2$$

Q4. If $\vec{a} = \vec{b} + \vec{c}, \vec{b} \times \vec{d} = 0$ and $\vec{c} \cdot \vec{d} = 0$, then $\frac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$ is equal to

- (A) $\vec{a}$ (B) $\vec{b}$ (C) $\vec{c}$ (D) $\vec{d}$

Ans. C

$$\text{Sol. } \vec{a} = \vec{b} + \vec{c}$$

$$\vec{a} \times \vec{d} = \vec{b} \times \vec{d} + \vec{c} \times \vec{d}$$

$$\vec{a} = \vec{b} + \vec{c}$$

$$\vec{a} \times \vec{d} = \vec{b} \times \vec{d} + \vec{c} \times \vec{d}$$

$$= \frac{\vec{d} \times (\vec{a} \times \vec{d})}{\vec{d}^2}$$

$$= \frac{\vec{d} \times (\vec{c} \times \vec{d})}{\vec{d}^2}$$

$$= \frac{\vec{d}(\vec{c} \times \vec{d})}{\vec{d}^2}$$

$$= \frac{\vec{c}(\vec{d}^2) - (\vec{c} \cdot \vec{d})\vec{d}}{\vec{d}^2}$$

$$= \vec{c}$$

Q5. The angles of a triangle, two of whose sides are represented by the vectors $\sqrt{3}(\hat{a} \times \vec{b})$ and $\vec{b} - (\hat{a} \cdot \vec{b})\hat{a}$, where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector in the direction of $\vec{a}$, are :

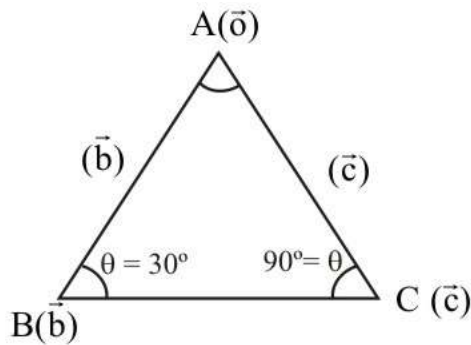
(A) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right); \tan^{-1}\left(\frac{1}{2}\right); \tan^{-1}\left(\frac{\sqrt{5}+2}{1-2\sqrt{3}}\right)$ (B) $\tan^{-1} \sqrt{3}; \tan^{-1}\left(\frac{1}{\sqrt{3}}\right); \cot^{-1}(0)$

(C) $\tan^{-1}(\sqrt{3}); \tan^{-1}(2); \tan^{-1}\left(\frac{\sqrt{3}+2}{2\sqrt{3}-1}\right)$

(D) $\tan^{-1}(\sqrt{3}); \tan^{-1}(\sqrt{2}); \tan^{-1}\left(\frac{\sqrt{2}+3}{3\sqrt{2}-1}\right)$

Ans. B

Sol.



Let $\vec{b} = \sqrt{3}(\hat{a} \times \hat{b})$

$$\vec{c} = \vec{b} - (\hat{a} \cdot \vec{b})\hat{a}$$

$$\vec{b} \cdot \vec{c} = \sqrt{3}(\vec{a} \times \vec{b}) \cdot (\vec{b} - (\vec{a} \cdot \vec{b})\hat{a})$$

$$= 0$$

sin rule

$$\frac{|\vec{c}|}{\sin \theta} = \frac{|\vec{b}|}{\sin (90 - \theta)}$$

$$\frac{|\vec{b} - (\hat{a} \cdot \vec{b})\hat{a}|}{\sin \theta} = \frac{\sqrt{3}|\hat{a} \times \hat{b}|}{\cos \theta}$$

$$\tan \theta = \frac{1}{\sqrt{3}} \frac{|\vec{b} - (\vec{a} \cdot \vec{b})\hat{a}|}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}} \frac{|(\vec{a} \cdot \vec{b}) \times \hat{a}|}{|\vec{a} \times \vec{b}|}$$

$$= \frac{1}{\sqrt{3}} \frac{|\hat{a} \cdot \vec{b}| |\hat{a}| \sin 90^\circ}{|\hat{a} \times \vec{b}|}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\angle A = 90^\circ$$

$$\angle B = 30^\circ$$

$$\angle C = 60^\circ$$

Q6. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ . Then $[\vec{a}\vec{b}\vec{c}]$ in terms of θ is equal to:

(A) $(1 + \cos \theta)\sqrt{\cos 2\theta}$ (B) $(1 + \cos \theta)\sqrt{1 - 2 \cos 2\theta}$ (C) $(1 - \cos \theta)\sqrt{1 + 2 \cos \theta}$

(D) None of these

Ans. C

Sol.

$$\begin{aligned} [\vec{a}\vec{b}\vec{c}]^2 &= \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} \\ &= \begin{bmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{bmatrix} \\ &= 1 - (1 - \cos^2 \theta) - \cos \theta (\cos \theta - \cos^2 \theta) + \cos \theta (\cos^2 \theta - \cos \theta) \\ &= 1 - \cos^2 \theta - \cos^2 \theta + \cos^3 \theta + \cos^3 \theta - \cos^2 \theta \\ &= 2 \cos^3 \theta - 3 \cos^2 \theta + 1 \\ [\vec{a}\vec{b}\vec{c}]^2 &= (1 - \cos \theta)^2 (1 + \cos \theta) \\ [\vec{a}\vec{b}\vec{c}] &= (1 - \cos \theta) \sqrt{1 + \cos \theta} \end{aligned}$$

Q7. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does NOT exceed

(A) 4 (B) 9 (C) 8 (D) 6

Ans. B

Sol. $\hat{a}, \hat{b}, \hat{c}$ are unit vector

$$\therefore \hat{a} \cdot \hat{a} = \hat{b} \cdot \hat{b} = \hat{c} \cdot \hat{c} = 1$$

$$\text{Now, } x = |\hat{a} - \hat{b}|^2 + |\hat{b} - \hat{c}|^2 + |\hat{c} - \hat{a}|^2$$

$$= \hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} - 2\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{b} + \hat{c} \cdot \hat{c} -$$

$$2\hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{c} + \hat{a} \cdot \hat{a} - 2\hat{c} \cdot \hat{a}$$

$$= 6 - (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \dots\dots\dots(1)$$

$$\text{Also } |\hat{a} + \hat{b} + \hat{c}| \geq 0$$

$$\Rightarrow \hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} + \hat{c} \cdot \hat{c} + 2(\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \geq 0$$

$$\Rightarrow 3 + 2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \geq 0$$

$$\Rightarrow -2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \leq 3$$

$$\Rightarrow 6 - 2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \leq 9 \quad \dots\dots\dots(2)$$

From (1) and (2), $x \leq 9$

$\therefore x$ does not exceed 9

Q8. Let $\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{W} = \hat{i} + 3\hat{k}$. If $\vec{U}$ is a unit vector, then the maximum value of the scalar triple product $[\vec{U}\vec{V}\vec{W}]$ is

- (A) -1 (B) $\sqrt{10} + \sqrt{6}$ (C) $\sqrt{59}$ (D) $\sqrt{60}$

Ans. C

Sol. $[\vec{u}\vec{v}\vec{w}] = \vec{u} \cdot (\vec{v} \times \vec{w})$

$$= |\vec{u}| |\vec{v} \times \vec{w}| \cos \theta$$

$$\vec{v} \times \vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix}$$

$$= \hat{i}(3) - \hat{j}(6 + 1) + \hat{k}(-1)$$

$$= 3\hat{i} - 7\hat{j} - \hat{k}$$

$$= |\vec{u}| |\vec{v} \times \vec{w}| \cos \theta$$

$$= (1) \sqrt{59} \cos \theta$$

$$[\vec{u} \vec{v} \vec{w}]_{\max} = \sqrt{59} (1)$$

$$= \sqrt{59}$$

Q9. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$, then $\vec{b} =$

- (A) $\hat{i}$ (B) $\hat{i} - \hat{j} + \hat{k}$ (C) $2\hat{j} - \hat{k}$ (D) $2\hat{i}$

Ans. A

Sol. Let $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ b_1 & b_2 & b_3 \end{vmatrix} = (b_3 - b_2)\hat{i} + (b_1 - b_3)\hat{j} + (b_2 - b_1)\hat{k}$$

$$= \hat{j} - \hat{k}$$

$$b_3 - b_2 = 0 \quad \dots\dots\dots(1)$$

$$b_1 - b_3 = 1, \quad \dots\dots\dots(2)$$

$$b_2 - b_1 = -1 \quad \dots\dots\dots(3)$$

$$\vec{a} \cdot \vec{b} = 1$$

$$b_1 + b_2 + b_3 \quad \dots\dots\dots(4)$$

from (1), (2), (4)

$$b_2 = 0 = b_3$$

$$b_1 = 1$$

$$\vec{b} = \hat{i}$$

Q10. Let $\vec{A}$ be vector parallel to line of intersection of planes P_1 & P_2 through origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ & $4\hat{j} - 3\hat{k}$ & P_2 is parallel to $\hat{j} - \hat{k}$ & $3\hat{i} + 3\hat{j}$, then the angle between vector $\vec{A}$ & $2\hat{i} + \hat{j} - 2\hat{k}$ is

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Ans. B

Sol. Normal to plane P_1 is

$$\vec{n}_1 = (2\hat{j} - 3\hat{k}) \times (4\hat{j} - 3\hat{k}) = -18\hat{i}$$

Normal to plane P_2 is

$$\vec{n}_2 = (\hat{j} - \hat{k}) \times (3\hat{i} + 3\hat{j}) = -3\hat{i} - 3\hat{j} - 3\hat{k}$$

$$\therefore \vec{A} \text{ is parallel to } (\vec{n}_1 \times \vec{n}_2) = (-54\hat{j} + 54\hat{k})$$

Now, angle between $\vec{A}$ and $2\hat{i} + \hat{j} - \hat{k}$ is given by $\cos \theta =$

$$\frac{(-54\hat{j} + 54\hat{k}) \cdot (2\hat{i} + \hat{j} - \hat{k})}{54\sqrt{2} \cdot 3} = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

Q11. A variable plane at a distance of 1 unit from the origin cuts the co-ordinate axes at A, B and C. If the centroid D (x, y, z) of triangle ABC satisfies the relation $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = k$, then the value of k is

- (A) 3 (B) 1 (C) 1/3 (D) 9

Ans. D

Sol. Let the equation of the plane ABCD be $ax + by + cz + d = 0$, the point A" be (α, β, γ) and the height of the parallelopiped ABCDA'B'C'D' be h.

$$\Rightarrow \frac{|a\alpha + b\beta + c\gamma + d|}{\sqrt{a^2 + b^2 + c^2}} = 90\% \cdot h$$

$$\Rightarrow a\alpha + b\beta + c\gamma + 0.94 \sqrt{a^2 + b^2 + c^2}$$

$$\therefore \text{locus is } ax + by + cz + d = 0.9h \sqrt{a^2 + b^2 + c^2}$$

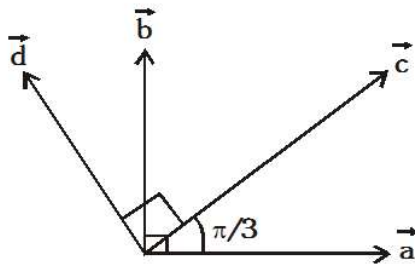
$\therefore$ locus of A" is a plane parallel to the plane ABCD.

Q12. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then

- (A) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar (B) $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar (C) $\vec{b}, \vec{d}$ are non-parallel
(D) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel

Ans. C

Sol.



$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1 \quad \dots\dots\dots(1)$$

$$\text{Let } \vec{a} \wedge \vec{b} = \alpha$$

$$\vec{a} \wedge \vec{b} = \beta$$

angle between plane of $(\vec{a}, \vec{b})$ & $(\vec{c}, \vec{d})$ be θ equation (1) becomes

$$\sin \alpha \cdot \sin \beta \cos \theta = 1$$

$$\Rightarrow \alpha = \frac{\pi}{2}, \beta = \frac{\pi}{2}, \theta = 0$$

$\Rightarrow \vec{b}$ & $\vec{d}$ are non-parallel.

Q13. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by

- (A) $\hat{i} - 3\hat{j} + 3\hat{k}$ (B) $-3\hat{i} - 3\hat{j} - \hat{k}$ (C) $3\hat{i} - \hat{j} + 3\hat{k}$ (D) $\hat{i} + 3\hat{j} - 3\hat{k}$

Ans. C

Sol. (a) $\vec{v} = x\vec{a} + y\vec{b}$

$$= \hat{i}(x+y) + \hat{j}(x-y) + \hat{k}(x+y) \quad \dots\dots\dots(i)$$

Given, $\vec{v} \cdot \vec{c} = \frac{1}{\sqrt{3}}$

$$\Rightarrow \frac{x+y-x+y-x-y}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$y-x=1$$

$$\Rightarrow x-y=-1 \quad \dots\dots\dots(ii)$$

using (ii) in (i) we get

$$\vec{v} = (x+y)\hat{i} - \hat{j} + (x+y)\hat{k}$$

Q14. If a, b, c are different real numbers and $a\hat{i} + b\hat{j} + c\hat{k}$, $b\hat{i} + c\hat{j} + a\hat{k}$ and $c\hat{i} + a\hat{j} + b\hat{k}$ are position vectors of three non-collinear points A, B and C, then

(A) centroid of triangle ABC is $\frac{a+b+c}{3}(\hat{i} + \hat{j} + \hat{k})$

(B) $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle.

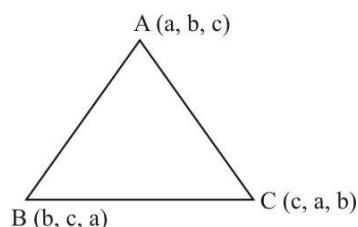
Ans. D, C, B, A

Sol. A (a, b, c) B (b, c, a) C (c, a, b)

(A) Centroid = $\frac{A+B+C}{3} = \left(\frac{a+b+c}{3}\right)(\hat{i} + \hat{j} + \hat{k})$

(B) $\cos \theta = \frac{a+b+c}{\sqrt{3}\sqrt{a^2+b^2+c^2}}$

(C)



Q15. If $\vec{z}_1 = a\hat{i} + b\hat{j}$ and $\vec{z}_2 = c\hat{i} + d\hat{j}$ are two vectors in $\hat{i}$ and $\hat{j}$ system, where $|\vec{z}_1| = |\vec{z}_2| = r$ and $\vec{z}_1 \cdot \vec{z}_2 = 0$, then $\vec{w}_1 = a\hat{i} + c\hat{j}$ and $\vec{w}_2 = b\hat{i} + d\hat{j}$ satisfy:

(A) $|\vec{w}_1| = r$ (B) $|\vec{w}_2| = r$ (C) $\vec{w}_1 \cdot \vec{w}_2 = 0$ (D) None of these

Ans. A, B, C

Sol. $Z_1 \cdot Z_2 = 0$ $|Z_1| = |Z_3| = r$

$$ac + bd = 0 \quad \sqrt{a^2 + b^2} = \sqrt{c^2 + d^2} = r$$

$$a^2 + b^2 = c^2 + d^2 = r^2$$

$$a = -\frac{bd}{c}$$

$$a^2 + b^2 = r^2$$

$$\left(\frac{bd}{c}\right)^2 + b^2 = r^2$$

$$(d^2 + c^2) \frac{b^2}{c^2} = r^2$$

$$r^2 \left(\frac{b^2}{c^2}\right) = r^2$$

$$b^2 = c^2 \quad \text{hence } a^2 = d^2$$

$$|w_1| = \sqrt{a^2 + c^2} = r$$

$$|w_2| = \sqrt{b^2 + d^2} = r$$

$$\vec{w}_2 \cdot \vec{w}_2 = ab + cd = ac + bd$$

$$= 0$$

Q16. If $\vec{a}, \vec{b}$, are two non-collinear unit vectors and $\vec{a}, \vec{b}, x\vec{a} - y\vec{b}$ form a triangle, then:

- (A) $x = -1; y = 1$ and $|\vec{a} + \vec{b}| = 2 \cos\left(\frac{\vec{a} \cdot \vec{b}}{2}\right)$
- (B) $x = -1; y = 1$ and $\cos\left(\frac{\vec{a} \cdot \vec{b}}{2}\right) + |\vec{a} + \vec{b}| \cos\left(\frac{\vec{a} \cdot (\vec{a} - \vec{b})}{2}\right) = -1$
- (C) $|\vec{a} + \vec{b}| = 2 \cot\left(\frac{\vec{a} \cdot \vec{b}}{2}\right) \cos\left(\frac{\vec{a} \cdot \vec{b}}{2}\right)$ and $x = -1, y = 1$ (D) none of these

Ans. B, A

Sol. $\vec{a}, \vec{b}, (x\vec{a} - y\vec{b})$ form a triangle

$$\vec{a} + \vec{b} + x\vec{a} - y\vec{b} = 0$$

$$\vec{a}(x+1) + (1-y)\vec{b} = 0$$

$$x+1=0, 1-y=0 \quad [\vec{a} \text{ \& } \vec{b} \text{ non collinear}]$$

$$x = -1, y = 1$$

let angle between (A) $\vec{a}$ & $\vec{b} = \theta$

$$|\vec{a} + \vec{b}| = \sqrt{a^2 + b^2 + 2ab \cos \theta} = \sqrt{2 + 2 \cos \theta} = 2 \cos \left(\frac{\theta}{2} \right)$$

$$(B) \cos \theta + |\vec{a} + \vec{a}| \cos \hat{a}$$

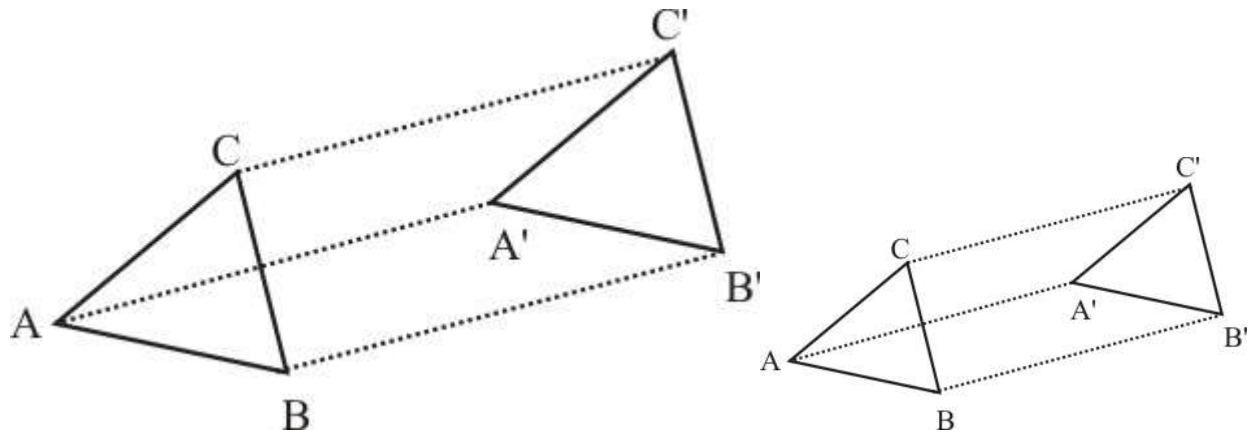
$$\cos \theta + 2 \cos \frac{\theta}{2}$$

Q17. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are $A(1, 0, 1)$, $B(2, 0, 0)$ and $C(0, 1, 0)$, then position vectors of the vertex A_1 can be:

- (A) $(2, 2, 2)$ (B) $(0, 2, 0)$ (C) $(0, -2, 2)$ (D) $(0, -2, 0)$

Ans. D, A

Sol.



$V = \text{Base area} \times \text{height}$

$$\text{Base area} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ -1 & 1 & -1 \end{vmatrix}$$

$$= \hat{i}(0 + 1) - \hat{j}(-1 - 1) + \hat{k}(1)$$

$$= \hat{i} + 2\hat{j} + \hat{k}$$

$$\text{Area} = \frac{1}{2} \sqrt{1 + 4 + 1} = \frac{\sqrt{6}}{2}$$

$$3 = \frac{\sqrt{6}}{2} \times h$$

$$h = \sqrt{6}$$

$$\text{check option } AA^1 = \sqrt{6}$$

Q18. Which of the following statement(s) is/are true?

(A) If $\vec{n} \cdot \vec{a} = 0$, $\vec{n} \cdot \vec{b} = 0$ and $\vec{n} \cdot \vec{c} = 0$ for some non-zero vector $\vec{n}$, then $[\vec{a}\vec{b}\vec{c}] = 0$.

(B) there exist a vector having direction angles $\alpha = 30^\circ$ and $\beta = 45^\circ$

- (C) locus of point for which $x = 3$ and $y = 4$ is a line parallel to the Z - axis whose distance from the Z -axis is 5
- (D) the vertices of a regular tetrahedron are $OABC$ where 'O' is the origin. Then vector $\vec{OA} + \vec{OB} + \vec{OC}$ is perpendicular to the plane ABC .

Ans. A, C, D

Sol. (A) $\vec{n} \cdot \vec{a} + \vec{n} \cdot \vec{b} + \vec{n} \cdot \vec{c} = 0$

$$\vec{n} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\vec{a} + \vec{b} + \vec{c} = 0 \quad \vec{a}, \vec{b}, \vec{c} \text{ are coplanar}$$

(B) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2 \gamma = 1$$

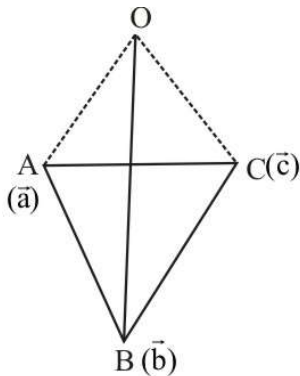
$$\frac{3}{4} + \frac{1}{2} + \cos^2 \gamma = 1$$

$$\cos^2 \gamma = \frac{-1}{4} \text{ (Not possible)}$$

(D) Normal = $\vec{AB} \times \vec{AC}$

$$= (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})$$

$$= \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$



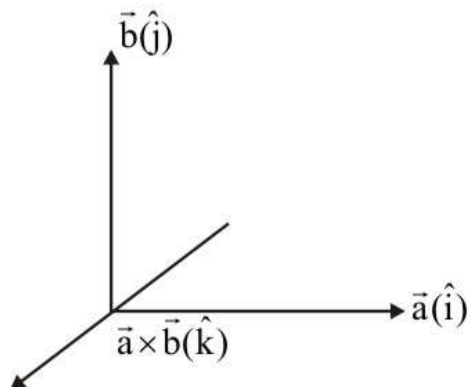
Q19. $\hat{a}$ and $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$, $\hat{b}$ and $\hat{a} \times \hat{b}$ will be:

(A) $-\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} + \hat{a} \times \hat{b})$ (B) $\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} + \hat{a} \times \hat{b})$ (C) $\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} - \hat{a} \times \hat{b})$

(D) $-\frac{1}{\sqrt{3}}(\hat{a} + \hat{b} - \hat{a} \times \hat{b})$

Ans. A, B

Sol.



Let new makes angle α with all axis

$$3 \cos^2 \alpha = 1$$

$$\cos \alpha = \pm \frac{1}{\sqrt{3}}$$

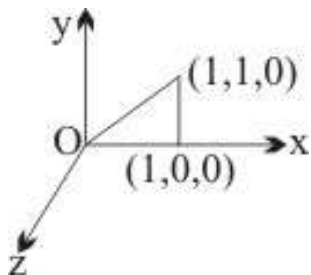
$$\begin{aligned} \text{vector} &= \pm \frac{1}{\sqrt{3}} (\hat{i}, \hat{j}, \hat{k},) \\ &= \pm \frac{1}{\sqrt{3}} (\hat{a}, \hat{b}, \hat{c}) \end{aligned}$$

Q20. Which of the following statement(s) is(are) incorrect?

- (A) The relation $|(\vec{u} \times \vec{v})| = |(\vec{u} \cdot \vec{v})|$ is only possible if atleast one of the vectors $\vec{u}$ and $\vec{v}$ is null vector.
- (B) Every vector contained in the line $\vec{r}(t) = \langle 1 + 2t, 1 + 3t, +1 + 4t \rangle$ is parallel to the vector $\langle 1, 1, 1 \rangle$.
- (C) If scalar triple product of three vectors $\vec{u}, \vec{v}, \vec{w}$ is larger than $|\vec{u} \times \vec{v}|$ then $|\vec{w}| > 1$.
- (D) The distance between the x-axis and the line $x = y = 1$ is $\sqrt{2}$.

Ans. D, B, A

Sol.



(A) Can be true if the angle between is 45°

(B) It is parallel to $(2, 3, 4)$

(C) Think of the scalar triple product as the volume. If that is larger than the base area, the height has to be bigger than 1.

(D) $x = y = 1$ denotes a line passing through (1, 1, 0) and parallel to z-axis.

Hence the distance between this line and x-axis is 1 and its distance from the z-axis is $\sqrt{2}$.]

Q21. Consider the family of planes $x + y + z = c$ where c is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations hold good for this family.

(A) each member of this family is equally inclined with the coordinate axes.

(B) $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$ (C) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units.

Ans. A, B, C

Sol. Vector normal to the plane

$$\vec{n} = \hat{i} + \hat{j}$$

$$\vec{V}_x = \hat{i}; \vec{V}_y = \hat{j}; \vec{V}_z = \hat{k}$$

$$\cos(90 - \alpha) = \frac{\vec{V}_x \cdot \vec{n}}{|\vec{n}|} \Rightarrow \sin \alpha = \frac{1}{\sqrt{3}}$$

$$\text{||y } \sin \beta = \frac{1}{\sqrt{3}} \text{ and } \sin \gamma = \frac{1}{\sqrt{3}}$$

$$\text{hence } \sum \sin^2 \alpha \text{ and } \sum \cos^2 \alpha = 2$$

Also plane is equally inclined with the coordinate axes.

$$\text{Also A} = \frac{1}{2} \sqrt{9^2 + 9^2 + 9^2} = \frac{9\sqrt{3}}{2} \Rightarrow \text{(D) is correct}$$

Q22. Which of the following statement(s) is/are true in respect of the lines

$$\vec{r} = \vec{a} + \lambda \vec{b}; \vec{r} = \vec{c} + \mu \vec{d} \text{ where } \vec{b} \times \vec{d} \neq 0$$

(A) acute angle between the lines is $\cos^{-1} \left(\frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}| |\vec{d}|} \right)$

(B) The lines would intersect if $[\vec{c} \vec{b} \vec{d}] = [\vec{a} \vec{b} \vec{d}]$.

(C) The lines will be skew if $[\vec{c} - \vec{a} \vec{b} \vec{d}] \neq 0$

(D) If the lines intersect at $\vec{r} = \vec{r}_0$ then the equation of the plane containing the lines is $[\vec{r} - \vec{r}_0 \vec{b} \vec{d}] = 0$.

Ans. A, B, C, D

Sol. (A) $\vec{r}(\vec{r} \cdot \hat{i})\hat{i} + (\vec{r} \cdot \hat{j})\hat{j} + (\vec{r} \cdot \hat{k})\hat{k}$

put $\vec{r} = (\vec{a} \times \vec{b}) \Rightarrow$ (A) is correct

(B) put $\vec{r} = \vec{a}$ and $\vec{r} = \vec{b}$

take $\vec{a} \cdot \vec{b} \Rightarrow$ (B) is correct

(C) $\vec{u} = \hat{b} \times (\hat{a} \times \hat{b})$

$|\vec{u}| = |\vec{a} \times \vec{b}| |\vec{V}| \Rightarrow$ (C) is correct

(D) is obviously wrong

Q23. Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct?

(A) $\vec{a} \times \vec{b} = \left[\begin{matrix} \vec{a} \cdot \hat{i} \\ \vec{a} \cdot \hat{j} \\ \vec{a} \cdot \hat{k} \end{matrix} \right] \hat{i} + \left[\begin{matrix} \vec{b} \cdot \hat{i} \\ \vec{b} \cdot \hat{j} \\ \vec{b} \cdot \hat{k} \end{matrix} \right] \hat{j} + \left[\begin{matrix} \vec{a} \cdot \hat{i} \\ \vec{a} \cdot \hat{j} \\ \vec{a} \cdot \hat{k} \end{matrix} \right] \hat{k}$ (B) $\vec{a} \times \vec{b} = (\vec{a} \cdot \hat{i})(\vec{b} \cdot \hat{i}) + (\vec{a} \cdot \hat{j})(\vec{b} \cdot \hat{j}) + (\vec{a} \cdot \hat{k})(\vec{b} \cdot \hat{k})$

(C) if $\vec{u} = \vec{a} - (\vec{a} \cdot \hat{b})\hat{b}$ and $\vec{v} = (\hat{a} \times \hat{b})$ then $|\vec{u}| = |\vec{v}|$

(D) if $\vec{c} = \vec{a} \times (\vec{a} \times \vec{b})$ and $\vec{d} = \vec{b} \times (\vec{a} \times \vec{b})$ then $\vec{c} + \vec{d} = \vec{0}$

Ans. A, B, C

Sol. (A) $\vec{r}(\vec{r} \cdot \hat{i})\hat{i} + (\vec{r} \cdot \hat{j})\hat{j} + (\vec{r} \cdot \hat{k})\hat{k}$

put $\vec{r} = (\vec{a} \times \vec{b}) \Rightarrow$ (A) is correct

(B) put $\vec{r} = \vec{a}$ and $\vec{r} = \vec{b}$

take $\vec{a} \cdot \vec{b} \Rightarrow$ (B) is correct

(C) $\vec{u} = \hat{b} \times (\hat{a} \times \hat{b})$

$|\vec{u}| = |\vec{a} \times \vec{b}| |\vec{V}| \Rightarrow$ (C) is correct

(D) is obviously wrong

Q24. The vector(s) which is/are coplanar with vectors $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$, and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ is/are

(A) $\hat{j} - \hat{k}$ (B) $-\hat{i} + \hat{j}$ (C) $\hat{i} - \hat{j}$ (D) $-\hat{j} + \hat{k}$

Ans. A, D

Sol. (b) $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$

$\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

$$\vec{c} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{v} = \lambda((\vec{a} \times \vec{b}) \times \vec{c}) = \lambda(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

$$\vec{v} = \lambda[4(\hat{i} + 2\hat{j} + \hat{k}) - 4(\hat{i} + \hat{j} + 2\hat{k})]$$

$$\vec{v} = 4\lambda(\hat{j} - \hat{k})$$